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for Information Security

Technical Guideline TR-03137

Optically Verifiable Cryptographic Protection of non-electronic Documents
(Digital Seal)

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Document history

Version	Date	Description
2.0	Apr 15, 2016	Merges part 1 and part 2 of BSI TR03137 v1.0 as well as the AAD profile v1.0
2.1	Feb 13, 2017	Integration of a Social Insurance Card Profile and adapting of length bytes in the profile to increase the allowed storage of data in the seal.
2.2	Sept 10, 2018	Harmonization with the requirements of ICAO TR Visible Digital Seals for Non-Electronic Documents Version 1.6
2.3	March 10, 2020	Integration of a Residence Permit Document profile. The digital seal encoding of this TR is now integrated into the [TR-SEAL] of the ICAO.
2.4	Apr 09, 2021	Integration of VISA document profile and a worked example according to [TR-SEAL].
2.5	Oct 26, 2021	Integration of address sticker profile for the German Identity card, the residence sticker profile for the German passport and the worked examples. Revising of References from [TR-SEAL] to ICAO Doc 9303 series.
3.0	Aug 04 , 2026	Integration of a generic profile according to [ICAO TR-IDB] and the corresponding document profiles. These profiles allow the encoding of face images in addition to the printed data of the document within the bar code to increase the protection of digital seals against falsifications. In addition, the profile for the address sticker for residence permit is defined.

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1 Introduction

Nowadays, many electronic documents are distributed including a cryptographic signature, which allows a reliable verification of the integrity and authenticity of the document. Even travel and ID documents are issued with embedded electronic storage media, equipped with radio-frequency (RF) chips or issued as smart cards in order to provide such a verification of their data. However, there are still non-electronic documents whose authenticity and integrity should be verifiable, such as non-electronic travel and ID documents, e.g. visas, and passports, or driver licenses, and breeder documents (i.e. documents that can serve as a basis to obtain other, more important identification documents).

The digital seal is a mechanism that allows to verify (in parts or as a whole) the authenticity and integrity of non-electronic documents automatically by cryptographic means using optical document readers only. The digital seal is a two-dimensional bar code printed on the document and comprises digitally signed data for checking the authenticity of the document and the integrity of relevant document contents, e.g. text or images, as well as optionally additional biometric features of the document holder in case of travel or identity documents. Thus, the digital seal provides a considerable security improvement for non-electronic (usually paper based) documents. Nevertheless, compared to chip based documents there are considerable limitations. Storage capacity of digital seals is usually limited to a few kilobytes at most and neither the data nor the cryptographic keys or schemes for the digital seal can be updated on existing documents. That is why cryptographic agility is not supported. The digital seal does not provide any protection against cloning, does not implement privacy protection functionality, and is more prone to read errors due to wear and tear than chip based documents.

Printing and optical reading of integral parts of the non-electronic documents is, of course, prone to inaccuracy. Error tolerant techniques which enable correcting tolerable deviations while reading and decoding document information should therefore be applied in using the digital seal.

In general the generation of a digital seal encompasses the following steps:

1. Create digital reference data assigned to the document (and if applicable biometric data of the document holder) which shall be stored in the digital seal for enabling an automatic verification.
2. Add auxiliary data if needed and necessary information on signature methods and algorithms used to protect the stored data.
3. Digitally sign the reference data and the auxiliary data to protect the document information stored in the digital seal and to prove the integrity and authenticity of the document information.
4. Encode the information (reference and auxiliary data, digital signature) in a machine-readable bar code.
5. Attach (print) the coded information to (on) the document.

This document is structured as follows: In chapter 2, an informative introduction is given that sketches out the whole use case. In chapter 3, two different encodings according to [DOC 9303-13] or [ICAO TR-IDB] of a digital seal are specified. The contents of a digital seal as well as the required public key infrastructure are highly dependent on the specific use case. Each use case can be described in a profile. The necessary contents of such a profile are defined in chapter 4. Actual profiles according to [DOC 9303-13] are defined in chapter 5, 6, 7, 8, 9 and 10. Chapter 11 describes a generic profile according to [ICAO TR-IDB] where the specific document profiles in chapter 12 to 17 are mapped to. These profiles are able to store face images in addition to the data printed on the document, within the digital seal. Chapter 18 gives a short informative description for the use case of encoding biometric data in the digital seal. Chapter 19 gives an informative description of additional possible use cases. Annex A: Conversion of ECDSA Signature Formats (Informative) gives an informal description on signature encoding for the seal. Annex B: C40-Encoding of Strings (Normative) describes the encoding method for alphanumeric strings defined in chapter 3. Annex C: Worked Example Arrival Attestation Document (Informative) gives a worked example based on the profile defined in chapter 6. Annex D: Worked Example of a Social Insurance Card shows an example based on the profile from chapter 7. Annex E: Worked Example of a Digital Seal for Residence Permit Documents gives an example for

digital seals of paper-based Residence Permit documents according to the defined profile in chapter 8. Annex F: Worked Example for a Digital Seal for VISA Stickers describes a worked example for VISA sticker. Annex G: Worked Example for the Address Sticker for the German ID-Document describes the worked example for the digital seal of the address sticker of the German ID card and Annex H: Worked Example for the Residence Sticker for German Passports describes the digital seal for residence stickers for the German passport. In Annex I: Worked Example for Temporary or Substitutional Travel and ID Documents one of an example is given, how data are encoded according to [ICAO TR-IDB].

1.1 Terminology and Definitions

The key words "MUST", "SHALL", "REQUIRED", "SHOULD", "RECOMMENDED", and "MAY" in this document are to be interpreted as described in [RFC2119].

MUST	This word, or the terms "REQUIRED" or "SHALL", mean that the definition is an absolute requirement of the specification.
MUST NOT	This phrase, or the phrase "SHALL NOT", mean that the definition is an absolute prohibition of the specification.
SHOULD	This word, or the adjective "RECOMMENDED", mean that there may exist valid reasons in particular circumstances to ignore a particular item, but the full implications must be understood and carefully weighed before choosing a different course.
SHOULD NOT	This phrase, or the phrase "NOT RECOMMENDED" mean that there may exist valid reasons in particular circumstances when the particular behavior is acceptable or even useful, but the full implications should be understood and the case carefully weighed before implementing any behavior described with this label.
MAY	This word, or the adjective "OPTIONAL", mean that an item is truly optional. One vendor may choose to include the item because a particular marketplace requires it or because the vendor feels that it enhances the product while another vendor may omit the same item. An implementation which does not include a particular option MUST be prepared to interoperate with another implementation which does include the option, though perhaps with reduced functionality. In the same vein an implementation which does include a particular option MUST be prepared to interoperate with another implementation which does not include the option (except, of course, for the feature the option provides.)

We identify binary values with their hexadecimal representation preceded by 0x, e.g. 0x2a. If clear from context, the prefix 0x is sometimes omitted. Sometimes binary values are identified with their decimal value, written by appending dec, i.e. 42dec. Throughout this document, we assume Big Endian encoding, i.e. byte sequences are read from left to right. Bit sequences are read from right to left, i.e. we assume the least significant bit (LSB) to be at (the rightmost) position 0.

Moreover we define the following terminology:

Affine

Two records of a biometric feature are affine if they are from the same part of the body and thus have the same origin.

Bar Code

Optical, machine-readable representation, in one or two dimensions, of data relating to the object to which it is attached.

Bar Code Symbology

A mapping between messages and bar codes is called a symbology. Such a mapping is defined in the specification of the bar code and includes the encoding of single digits or characters, the size of a so called quiet zone around the bar code, as well as the computation of checksums for error correction.

Certificate

An electronic file attesting that a cryptographic key pair belongs to a person or a hardware or software component as identified in the certificate. A certificate is issued by a Certification Authority. By signing the certificate, the Certification Authority approves the link between the identity of a person or component and the cryptographic key pair. The certificate may be revoked if it doesn't attest the validity of this link any more. The certificate has a limited validity period.

Certificate Revocation List (CRL)

A list of certificates that have been revoked. Documents that identify a certificate from a CRL for verification shall thus no longer be trusted.

Country Signing Certificate Authority (CSCA)

The Certification Authority of a country that signs document signer certificates. Document issuers, such as manufacturers of passports, use the private keys corresponding to the document signer certificates to sign data on electronic machine readable travel documents (eMRTDs). The CSCA of each Issuing State or organization acts as the trust point for the receiving state. The CSCA also signs SubCA-Certificates for subordinate Certification Authorities (SubCAs) below the CSCA.

Cryptographic Signature

The output generated by a signature algorithm of a signature scheme. The signature ensures integrity and authenticity of data.

Cryptographic Signature Scheme

A tuple of three algorithms. The key-generation algorithm takes as input a security parameter and outputs a key pair consisting of a private and a public key. The signature algorithm takes as input a private key and a message, and outputs a cryptographic signature. The verification algorithm takes as input a public key, a message and a signature, and returns valid if the signature was generated using the signature generation algorithm with the private key of the key pair and the message as input, and invalid otherwise.

(Digital) Document Feature

A property of a document which can be used to verify the contents of the document or tie the document to its holder. Examples are textual information such as the name of the holder, the issuing date or a printed image of the document holder. A digital document feature is the digitized version of a document feature.

Digital Seal

A two-dimensional bar code printed on the document that contains digitally signed data for checking the authenticity and integrity of relevant document contents, e.g. text or images, as well as optionally additional biometric features of the document holder.

Elliptic Curve Digital Signature Algorithm (ECDSA)

A variant of the Digital Signature Algorithm (DSA) based on elliptic curve cryptography.

False-Acceptance-Rate (FAR)

A FAR describes the probability of non-affine biometric features which are wrongly recognized as a biometric match.

False-Rejection-Rate (FRR)

A FRR describes the probability of affine biometric features wrongly not recognized as a biometric match.

Machine Readable Travel Document (MRTD)

A travel document as defined in [MRTD6].

Machine Readable Zone (MRZ)

A fixed dimensional area on the MRTD consisting of textual data printed in a font designed for easy Optical Character Recognition (OCR). There are different types of MRZ, i.e. MRZ-TD1 (see [MRTD5]), MRZ-TD2 (see [MRTD6]), MRZ-TD3 (see [MRTD4]).

Physical Document Features

Physical properties of a document that prevent forging or faking it. Examples are watermarks, holograms or micro-printing.

Signature Scheme

See cryptographic signature scheme.

Signer

The authority that receives data from a document personalization system and that uses a certificate and the corresponding private key to encode and sign a digital seal.

Signer Certificate

A certificate containing information identifying the entity that signed a digital seal on a document containing the public key corresponding to the private key with which the signature was created.

(Feature) Tag

A byte that uniquely identifies a document feature. The mapping between feature tags and features must be specified in a profile.

Verifying Party

The authority that validates a visible digital seal based on a validation policy.

2 General Description (Informative)

This section gives a general overview of using a digital seal to protect a non-electronic document. The general use case is depicted in Figure 1. Technical details vary from use case to use case, but the same general principles apply.

The general workflow can be separated into three steps. As a prerequisite, Signer Certificates have to be generated. Next, seals are generated, and then later validated.

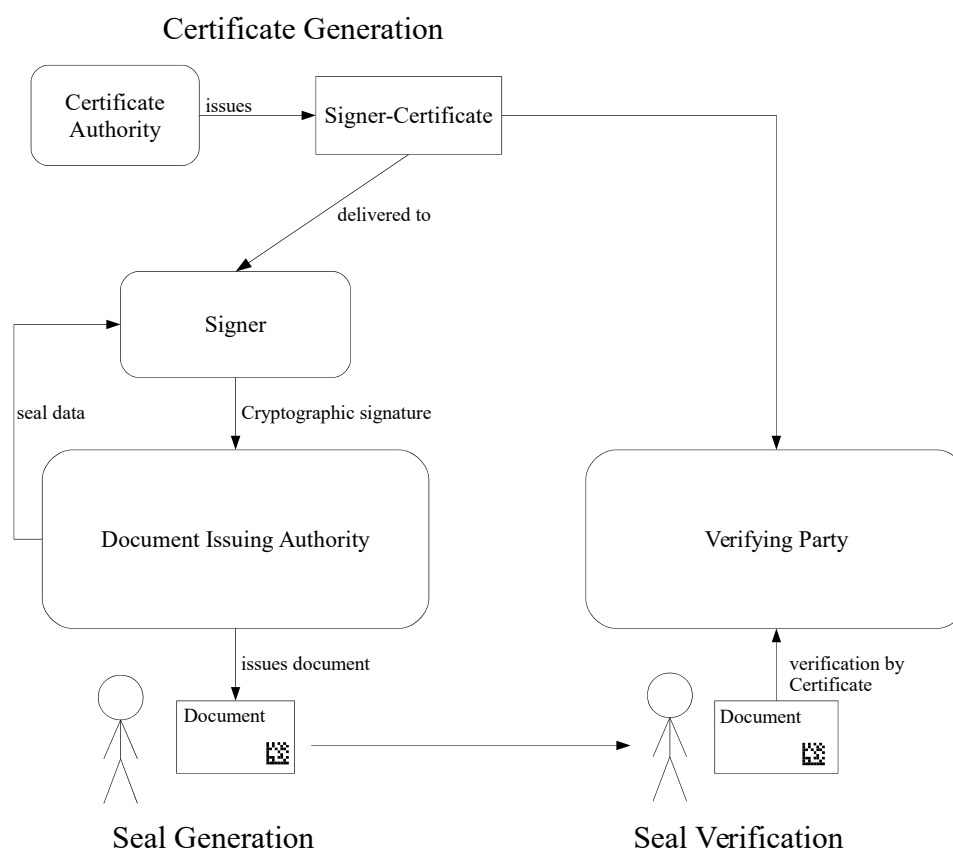


Figure 1: Use case

2.1 Signer Certificate Generation

The signer is the entity that actually signs **digital seals**. The signer uses a private key to create the signature for the seals. For the corresponding public key, a certificate is issued by a certificate authority (CA), and the CA publishes the certificate. For a document containing a digital seal, the Signer Certificate can be used to check the validity of a digital seal. Note that in practice, there can be one or more subordinate CA(s) below the (root) CA, and the distribution mechanism for certificates from the CA to the verifying party can vary. The whole process of seal generation and validation is described in the next two sections.

2.2 Digital Seal Generation

The number and type of features included in a digital seal may differ, depending on the specific use case. Those document features stored in the digital seal enable checking that the document has not been manipulated or forged as well as the readout of feature data in cases where information printed on the document is damaged. Additional biometric features (e.g. protected fingerprint templates) stored in the digital seal enable a reliable identification of the document holder. A digital signature of all document and additional biometric features stored in the digital seal protects the authenticity and integrity of the verification data.

A seal is generated in two steps:

1. An applicant applies for a document at the document issuing authority. The issuing authority records the applicant's data and checks whether the applicant meets the requirements to receive the document. If the requirements are fulfilled, the issuing authority sends a digital representation of the recorded data to the signer. The signer cryptographically signs the recorded data.
2. For signing, the signer uses the private key for which the corresponding public key was published by the CA as a certificate. The resulting signature is sent back to the issuing authority, printed on the document, and the document can be handed to the applicant.

2.3 Digital Seal Validation

Suppose the applicant's document is to be verified by a verifying party. The verifying party checks the authenticity and integrity of the digital seal on the document by validating the signature of the seal, and comparing the printed information on the document with the digital information stored in the seal. The signature of the seal is verified by identifying the corresponding signer certificate with the help of the identifier stored in the header of the seal, and then using the public key of the signer certificate.

Further Information

A digital seal is a data structure stored in a two-dimensional bar code printed on non-electronic documents (e.g. temporary identification or travel documents). It allows the verification of the authenticity and the integrity of the document contents as well as (optionally) a biometric verification of its holder. For this purpose the digital seal comprises digitally signed data by which the authenticity and integrity of features, which is either text printed on a document, biometric data itself (e.g. fingerprints) or a secure biometric template, can be verified. The digital seal is generated during the issuance of the document. Therefore, relevant data of the features to be covered by the seal are compiled. The issuing authority (or an entity acting on behalf it) signs the data digitally. Finally, the feature data and a digital signature are encoded and printed as a bar code on the document.

Document features and/or additional biometric features can be either read (and decoded) from the digital seal, captured by optical reading (scanning) of document elements, or be acquired by other means. These features may cover, but are not limited to, the following types:

1. Text printed on the document that can be read by optical means, e.g. content of the machine-readable zone (MRZ), the holder's identity information or serial number of the document.
2. A facial image of the document holder printed on the document.
3. Metadata of the document that cannot or do not need to be captured by optical reading, e.g. document type or the number of pages.
4. Other security features specified for the document, e.g. watermarks, microprinting, holograms, UV/IR-visible patterns or substance features.

5. A specimen handwritten signature of the document holder printed on the document.
6. Additional generated biometric features of the document holder, for instance fingerprints or iris patterns.

All features which are an **integral part** of the document, e.g. printed on the document, and to be protected by cryptographic means (digital signature) against falsification are considered (and denominated) in the following as **integral document features**. Biometric information that is not an integral part of the document but allows the identification of the document holder by acquisition of a biometric pattern are called **biometric document features**.

Some of the features which are printed on the document can also be used for biometric verification, e. g. facial images, images of a specimen signature or fingerprints, are henceforth referred to as document features too.

The digital seal is read optically, e.g. by scanning the document or using optical document readers. The authenticity and integrity of the digital seal is protected by a digital signature which can be verified using the public key of the signer (issuer of the document). The public key is referenced in the header data block of the digital seal (see section 3.2.1).

For each feature, the digital seal contains a data block called the **document features**, which depend on the content and which are either the printed text of the document, a biometric image or a biometric template. These verification data encompass the actual feature data as well as optional auxiliary data for supporting the automatic verification. The verification data stored in the digital seal must allow a sufficiently error tolerant verification of the document features to ensure that documents exposed to aging, wear and tear, and damages typically tolerated for that document type can still be successfully verified.

Storing biometric data can lead to capacity issues of the bar code due to the data volume requirements of biometric features and/or templates themselves and because they must be enlarged with error-correction-codes. At the same time, the digital seal must ensure that unauthorized modifications of the document features or the digital seal itself are detected during verification.

In addition, document features can also be verified by other (possibly non-automated) means; examples include non-optical features like the number of pages of the document, tactile features or optical features like watermarks, guilloches or holograms if no corresponding sensors are available.

3 Digital Seal Encoding

A digital seal is a cryptographically signed data structure containing document features, encoded as a 2D bar code and printed on a document. The encoding of the digital seal depends on the document type and is either according to [DOC 9303-13]¹ or [ICAO TR-IDB].

3.1 Symbology and Print Requirements

The bar code SHOULD be printed in a way that reader equipment (i.e. off-the-shelf smartphones or scanners) are capable to reliably decode the bar code. Printing and scanning quality requirements depend on the document and application scenario, and MAY be specified in a profile that applies for the use case. Due to the fact that the quality of printing and scanning affects error rates and influences the robustness of digital seal verification, these requirements SHOULD ensure that the bar code (e.g. QR-Code, DataMatrix ECC 200 or Aztec-Code) containing the digital seal and all mandatory document features can be reliably verified. Another important requirement addresses symbol contrast of the bar code, because the digital seal might be printed on security paper with a colored background (e.g. green).

For assessing print quality and the capability to reliably decode the bar code, [PRINT] MUST be taken into account. When using standard inkjet printers, the bar code MUST be printed with a module size (size of one block of a 2D bar code) of at least 0.3386mm width/height per module, corresponding to 4 dots per module side length (i.e. 16 dots per module) on a 300 dpi printer, or 8 dots per module side length (i.e. 64 dots per module) on a 600 dpi printer. Deviations to the aforementioned requirements are only allowed if defined in the specific document profiles.

3.2 Encoding according to [DOC 9303-13]

The encoded bar code consists of a header, the message zone and the signature zone as defined in [DOC 9303-13]

It has to be noted that there are two different length encodings (for the document features in the message zone and the signature), depending on the Version Identifier in the header. The message zone MUST contain at least one document feature element, and a valid signature MUST always be present.

Figure 2 gives a general overview how a digital seal is encoded according to this specification into a DataMatrix ECC 200 code. Different encodings are used for the input data depending on their format, as it can be alphanumeric, a date or binary. An alphanumeric string is first encoded with the C40 encoding to binary data, and dates are encoded in a specific date format (see section 3.2.3) as binary data. After that, such binary data (including other raw binary data) is encoded using Base256Encoding (see [DTMX]) to DataMatrixECC 200.

1 Documents already in use such as: Visa, Arrival Attestation and Address-Sticker MUST use the encoding of the digital seal as defined in [DOC 9303-13]. A switching to the encoding of digital seals according [ICAO TR-IDB] for the arrival attestation document will be done after a defined overlap period.

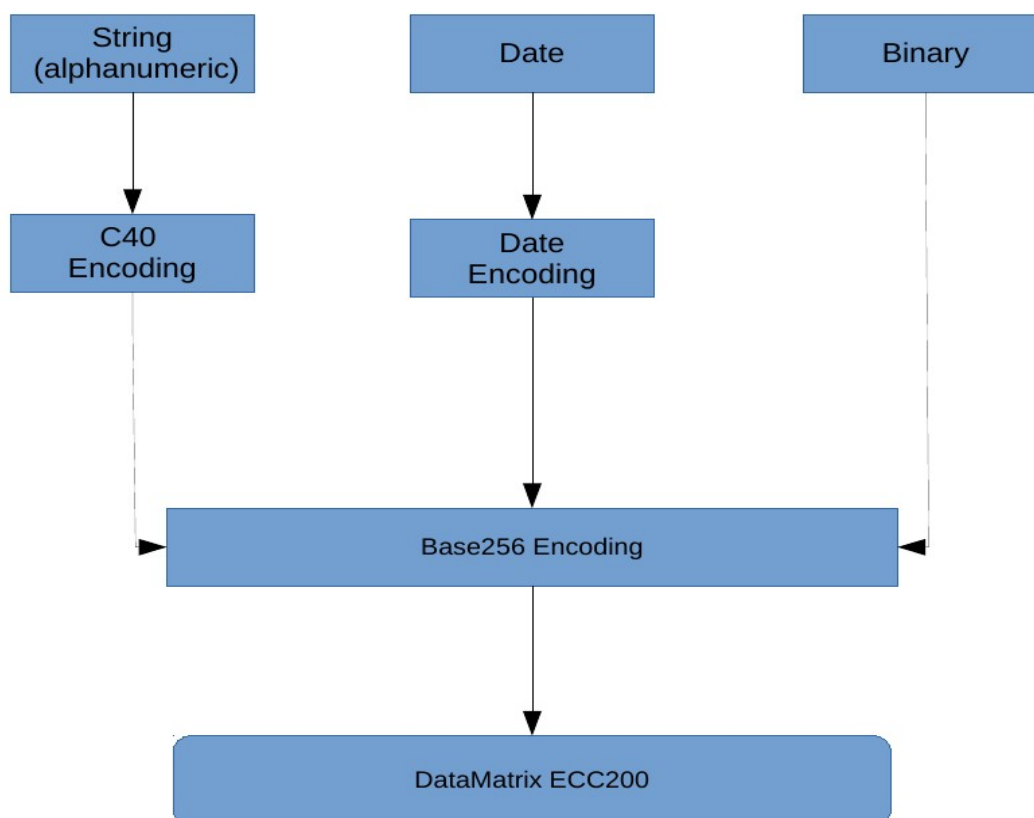


Figure 2: Encoding process of DataMatrix ECC 200 bar code

3.2.1 Header

The encoding of the header for the digital seal, which SHALL be used in all profiles of this TR, is defined in [DOC 9303-13]. In addition, the requirements defined in section 3.1.1 of [DOC 9303-13] applies.

3.2.2 Message Zone

The message zone encoding MUST be according to [DOC 9303-13].

3.2.3 Digital Encoding of Document Features

As building blocks to encode document features, three basic types are supported: *Alphanum* are strings of uppercase alphanumeric characters (i.e. A-Z, 0-9 and space), *Int* are positive integers, and *Date* is to store dates. These are converted to sequences of bytes as follows:

1. Strings of alphanumeric characters are encoded as bytes by C40 encoding (cf. Annex B: C40-Encoding of Strings (Normative)).
2. For positive integers, their unsigned integer representation is taken.
3. A date is first converted into a positive integer by concatenating the month, the days, and the (four digit) year. This positive integer is then concatenated into a sequence of three bytes (see [DOC 9303-13]).
4. *Binary* are sequences of bytes which are taken as they are.

If the three basic types above are unsuitable to encode the desired document feature (for example, if the document feature requires non-ASCII characters), the specific encoding of that document feature into *binary* MUST be defined in the specific profile. Otherwise, the above encoding MUST be used.

Example

Consider March 25th, 1957. Concatenating the month, date and year yields the integer 03251957, resulting in the three bytes 0x31 0x9e 0xf5.

A digital document feature is a sequence of bytes. It has the structure: tag | length | value.

Here tag is a unique integer in the range 0-254 acting as an identifier of the document feature, value is a basic type converted to a sequence of bytes, and length is an integer in the range 0-254 denoting the length (the number of bytes) of the value. Note that tag 255 is reserved to denote the start of the signature.

Example

Consider a document feature that encodes the string “BSI01” with assigned tag 0x0a. The C40 encoded byte sequence (cf. Section Annex B: C40-Encoding of Strings (Normative)) of length 4 is 0x62d719c9. The document feature is thus the byte sequence 0x0a0462d719c9.

Additional features, i.e. features with unknown tags MAY be present, for example for optional use of the issuing entity. Such additional features MUST NOT use the tag of the additional feature field, or the tag of any other optional or mandatory feature. The presence of features with unknown tags SHALL NOT affect the validity of the visa if the signature is recognized as valid.

3.2.4 Signature Zone

The signature zone is encoded as described in [DOC 9303-13].

3.2.5 Padding

If the header, message and signature together do not fill the available space of the bar code, padding characters shall be appended after the signature according to [DTMX].

3.3 Encoding according [ICAO TR-IDB]

The encoding of the bar codes follows the bar code structure as defined in [ICAO TR-IDB] section 3.1. This means the bar codes are consisting of a Bar Code Identifier, Bar Code Flag and the Bar Code Payload itself. The Bar Code Payload is defined in [ICAO TR-IDB] and is composed of:

- a header (see [ICAO TR-IDB] section 3.2),
- a message zone (see [ICAO TR-IDB] section 3.3),
- a signer certificate zone (see [ICAO TR-IDB] section 3.4) (optional)
- and signature zone ([ICAO TR-IDB] section 3.5).

Thus, the data encoded in the bar code are the concatenation of the Bar Code Identifier (composed of 4 characters), Bar Code Flag (1 character) and the Bar Code Payload (encoded as Base32). Figure 3 gives an overview how the data are encoded in to the bar code according to the [ICAO TR-IDB] specification.

The Bar Code Identifier MUST consists of exactly 4 characters by concatenating of “RDB” and the version by character “1”. All bar codes conforming to this technical guideline using [ICAO TR-IDB] encoding MUST be signed and MUST NOT be compressed. This means only value “B” is allowed for the Bar Code Flag, as described in [ICAO TR-IDB] section 3.1.2. [NOTE: This TR contains the Bar Code Identifier “RDB” instead of “NDB” in deviation to the current version of [ICAO TR-IDB]. The future version of [ICAO TR-IDB] will also contain the Bar Code Identifier “RDB”].

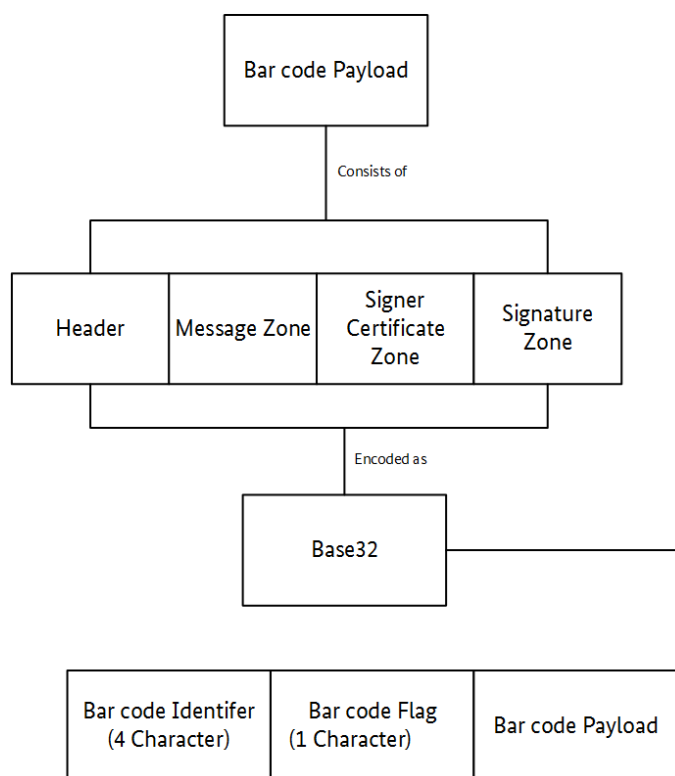


Figure 3: Overview of encoding a digital seal according to [ICAO TR-IDB]

Chapter 11 describes the general encoding of the bar code payload used for digital seal encoding as defined in [ICAO TR-IDB] section 3.1.4. The assignment of the different message types of the message zone is defined in the corresponding document profile.

4 Profile Definitions

A profile **MUST** contain definitions for the following aspects:

- restriction and formats of the bar code (i.e. size, capacity)
- the document features stored in the message zone of the digital seal including their tags and lengths
- the cryptographic signature format, algorithms, and key requirements
- the public key infrastructure
- certificate distribution
- certificate profiles for the signing certificates and other certificates of the public key infrastructure. The restrictions defined in Section 3.2.1 Signer Identifier and Certificate Reference (for bar codes according [DOC 9303-13]) **MUST** be taken into account when defining the certificate profiles
- validation of a digital seal on the document for the specific use case
- other necessary requirements specific to the use case, including custom encodings of document features into binary sequences

Every profile **MUST** adhere to the seal encoding defined in chapter 3. The ICAO-PKI as defined for digital seals [MRTD12] **SHOULD** be used whenever the specific use case allows it.

5 Visa Profile

This chapter specifies the profile for digital seals of VISA stickers issued by the German member state according to [C(2020) 2672]. VISAs are issued to persons entering Germany from a third-level country (neither an EU member state nor an EEA state). In addition to the personal data, the digital seal is printed on the VISA to ensure the integrity and authenticity of the document and to protect the document against falsifications. Thus, the digital seal increases the security of the VISA sticker by using cryptographic means.

5.1 Bar Code Symbology

The digital seal **MUST** be encoded as DataMatrix [DTMX] bar code with 44x44 modules (i.e. the maximum raw data capacity is 142 Bytes (see [DTMX])). The bar code **MUST** be printed with 600dpi with 7 dots per module side length (i.e. 49 dots per module). A Worked Example for digital seals for VISA stickers is given in Annex F: Worked Example for a Digital Seal for VISA Stickers.

5.2 Content and Encoding Rules

5.2.1 Header

The Version (at start position 0x01) in the header **MUST** be 4, i.e. 0x03 according to [DOC 9303-13] and the total length of the header **MUST** be 18 bytes. All other parameter of the header **MUST** be according to section 9.1.1 of [MRZ-VISA].

5.2.2 Document Features of the Digital Seal for VISA Stickers

The digital seal **SHALL** encode the following mandatory document features. Other document feature **MUST NOT** be present.

Machine Readable Zone (REQUIRED)

The Machine Readable Zone (MRZ) of VISA **SHALL** contain the following information (see [MRZ-VISA]):

- type of document
- issuing state
- surname and first name of the document holder
- passport or visa number
- nationality of the document holder
- date of birth of the document holder
- sex of the document holder
- validity period (valid until)

The information mentioned above **SHALL** be encoded as MRZ of MRV-B.

Duration of Stay (REQUIRED):

This document feature specifies the number of days, months or years the VISA holder may stay in the territory.

Passport Number (REQUIRED)

This document feature denotes the number of the passport to which the VISA is attached. The passport number is also required if it is already part of the MRZ of the VISA.

5.2.3 Encoding Rules for the Document Features

This section defines the concrete digital encoding of the aforementioned document features for VISA.

MRZ of Machine Readable VISA of Type B (according to [MRZ-VISA])

The encoding for the document feature MRZ of Machine Readable VISA of Type B SHALL be according to section 9.1.3 of [MRZ-VISA].

Duration of Stay

The document feature Duration of Stay SHALL be encoded as defined in section 9.1.3 of [MRZ-VISA].

Passport Number

The passport number of the passport on which the VISA is attached SHALL be encoded as defined in section 9.1.3 of [MRZ-VISA].

5.2.4 Signature

Hash and signature algorithms that SHALL be used for seal generation are defined in [KRYPT].

5.3 Public Key Infrastructure

For verifying the signature of the digital seal for VISA stickers, a VISA-Signer certificate is required. The VISA Signer certificate SHALL be issued by the CSCA. Thus the VISA-Signer certificates are issued in a way that allows verification by the CSCA certificate (defined in [MRTD12]). This means the VISA-Signer certificate is directly issued by the CSCA without having a SubCA in between. The VISA-Signer shall only use its private key for signing the data stored in the digital seal of VISA stickers. The corresponding public key is part of the VISA-Signer certificate, which is used to verify the signature on the digital seal.

In the sections below we use the following terminology for presence requirements of each of the components/extensions in certificates:

- m mandatory – the field MUST be present
- x do not use – the field MUST NOT be populated
- o optional – the field MAY be present

For the criticality of certificate extensions we use the following terminology:

- c critical – the extension is marked critical, receiving applications must be able to process this extension
- nc the extension is marked non-critical, receiving applications that do not understand this extension MUST ignore it

5.3.1 Certificates

CSCA Certificate

The profile for the CSCA certificate is defined in [MRTD12].

VISA-Signer Certificate Profile

The certificate profile for VISA-Signer Certificates is defined in the following Table 1.

Table 1: VISA-Signer Certificate Profile

Certificate Component	Presence	Remark
Certificate	m	
TBSCertificate	m	see below
signatureAlgorithm	m	dependent on selected algorithm
signatureValue	m	dependent on selected algorithm
TBSCertificate		
version	m	2 (indicating version 3)
serialNumber	m	MUST be the positive integer that results from interpreting the up to <i>three character hex-string</i> as a positive integer. This integer uniquely identifies a VISA-Signer Certificate for one CA and subject. leading bit must be zero in DER encoding (cf. [MRTD12]).
signature	m	this value MUST be the same as that in the signatureAlgorithm component of the certificate sequence
issuer	m	MUST be the value of the subject DN of the CSCA certificate with which the VISA-Signer certificate was signed
validity	m	MUST terminate with Zulu (Z). The seconds element MUST be present. Dates through 2049 must be in UTCTime, represented as YYMMDDHHMMSSZ. Dates in 2050 and beyond must be in GeneralizedTime. GeneralizedTime must not have fractional seconds, and must be represented as YYYYMMDDHHMMSSZ. The validity (i.e. difference between notBefore and notAfter) must be according to Section 5.3.2
subject	m	the following two MUST be present; other attributes MUST NOT be present. commonName: MUST consist of <i>two uppercase characters</i> in printableString format, that uniquely define the VISA-Signer within the issuing country, and must be present in the header of a VISA-Signer as the <i>Signer reference</i> countryName: MUST consist of the ISO two-letter country code of the VISA-Signer in uppercase characters, printableString format
subjectPublicKeyInfo	m	MUST adhere to [X509] and [MRTD12]

Certificate Component	Presence	Remark
issuerUniqueID	x	
subjectUniqueID	x	
Extensions	m	All mandatory extensions according to [MRTD12] SHALL be used. In addition, the DocumentType extension according to [MRTD12] MUST also be present.

5.3.2 Certificate Validity Periods

CSCA Certificate

The same validity periods as defined in section 6.4.3 SHALL be used.

VISA-Signer

Private Key Usage Time: 1 year + 2 months (the 2 months are to allow for smooth roll-over)

Certificate Validity: Private Key Usage Time + VISA Sticker Validity Timeframe

5.3.3 Certificate Publication

The publication of the VISA-Signer certificates MUST be done as described in section 6.5.3.

5.3.4 Certificate Revocation

If the private key of the VISA-Signer is compromised, the certificate SHALL be revoked. Requirements for the certificate revocation are described in [MRTD12].

6 Arrival Attestation Document Profile

This chapter specifies the profile for digital seals of the Arrival Attestation Document. The Arrival Attestation Document is a physical document which is surrendered to each refugee coming to Germany. This document contains a digital seal in addition to personal data to guarantee the authenticity of the document and for protection against forgery. In addition to personal data, biometric data are stored in a database to guarantee that the document is used by the correct person.

6.1 Bar Code Symbology

The seal **MUST** be encoded as a DataMatrix [DTMX] bar code with 48x48 blocks (max. 172 Byte raw capacity). For a worked example, see Annex C: Worked Example Arrival Attestation Document (Informative).

6.2 Content and Encoding Rules

6.2.1 Header

The *Document Feature Definition Reference* for this use case is 0xFD.

The *Document Type Category* for Arrival Attestation Documents is 0x02.

The version **MUST** be set to 0x02 according to [DOC 9303-13].

Otherwise, the content of the header is the same as defined in [DOC 9303-13].

6.2.2 Document Features of a Seal for an Arrival Attestation Document

The following document features are stored in the seal:

Machine Readable Zone (REQUIRED)

Basic Information is encoded using a Machine Readable Zone (MRZ) of a TD2-Format MROTD [MRTD6]. This MRZ contains the following information:

- document code
- issuing state
- surname and first name of the document holder
- document number
- nationality of the document holder
- date of birth of the document holder
- sex of the document holder
- date of expiry

Additionally, the following document features are stored:

AZR Number (REQUIRED)

A string consisting of 12 numeric characters.

6.2.3 Encoding Rules for Document Features

In the following, the digital encoding of document features of the arrival attestation document seal is defined.

MRZ (TD2-Type [MRTD6])

Tag: 0x02

Min. Length: 48 bytes

Max. Length: 48 bytes

Value Type: Alphanumeric

Presence: REQUIRED

Content: The first line and second line of the MRZ of a TD2-MROTD (2*36 chars.) MUST be concatenated and encoded by C40. The filler symbol < in the MRZ is replaced by <SPACE> prior to encoding by C40.

AZR Number

Tag: 0x03

Min. Length: 8 bytes

Max. Length: 8 bytes

Value Type: Numeric

Presence: REQUIRED

Content: The AZR Number: A numeric string of 12 characters, C40-encoded.

6.2.4 Signature

Hashing and signature algorithms defined in [KRYPT] for digital seals MUST be used.

6.3 Public Key Infrastructure

With regard to this profile for arrival attestation documents (AAD), AAD-Signer Certificates are issued in a way that allows for verification by CSCA certificates. This means there is no SubCA but only two entities: The AAD-Signer and the CSCA:

The AAD-Signer is the authority that signs AADs using a secret private key. The corresponding public key is contained in an AAD-Signer Certificate.

The CSCA according to [MRTD12] self issues CSCA certificates and is the root of the PKI. The CSCA also issues AAD-Signer Certificates.

As a consequence, requirements applying to AAD-Signer Certificates are aligned with regard to [MRTD12]. Note that the present specification defines additional requirements.

6.4 Seal Creation and Signing

6.4.1 Architecture of the Arrival Attestation Document Signer System

The AAD-Signer receives data from an AAD-Personalization System to encode a digital seal, and uses a signing key to sign it. Figure 4: AAD-Document personalization depicts this implementation of the AAD-Signer and its client, the AAD-Personalization System.

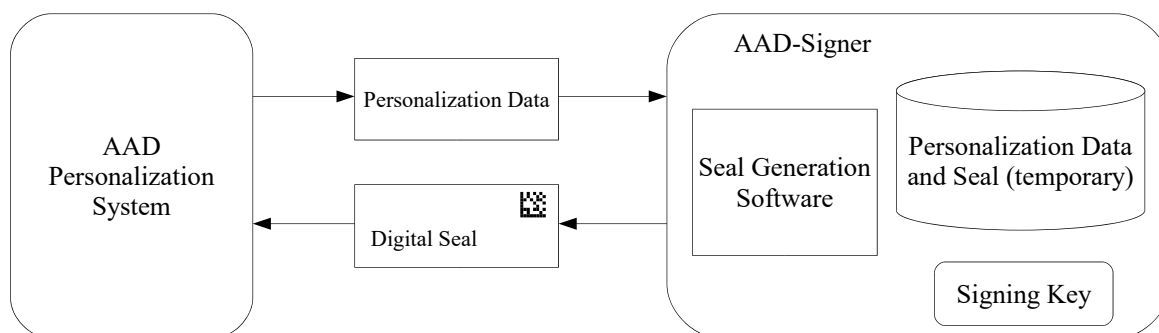


Figure 4: AAD-Document personalization

The AAD-Signer relies on the following software and data:

- The *seal generation software*, producing **digital seals** conforming to the present standard. It receives the personalization data sent by the client, signs these data with a private signing key, and encodes the personalization data and the signature to a bar code. The arrival attestation document personalization data and the seal are the input and output data of the seal generation software, and must be stored temporarily by the AAD-Signer during the generation of the seal.
- The *signature keys* (private and public key) to sign a **digital seal**. The private signing key is used by the seal generation software to sign the data of the seal. It is the most critical data of the AAD-Signer.

6.4.2 Management of Signature Keys

The signature keys are the most critical data of the AAD-Signer and as such SHALL be protected according to the best security practices.

6.4.3 Key Requirements (Validity Period)

Validity periods are as follows:

CSCA Certificates (as specified in [MRTD12])

Private Key Usage Time: 3 years

Certificate Validity: Private Key Usage Time + Max. of Key Lifetime (= Certificate Validity) of Document Signer Certificates, AAD-Signer Certificates or other certificates below the CSCA- whichever is longer

AAD-Signer Certificates

Private Key Usage Time: 1 year + 2 months (the 2 months are to allow for smooth roll-over)

Certificate Validity: Private Key Usage Time + AAD Validity Timeframe

The used hash function, signature algorithm and (domain) parameters must be present in the `SubjectPublicKeyInfo` extension of the AAD-Signer Certificate (cf. Section 6.5.2).

6.5 Certificates

The AAD-Signer Certificates shall be distributed using the CSCA-PKI [MRTD12]. A general outline of this procedure is briefly given in Section 2.1.

In the sections below we use the following terminology for presence requirements of each of the components/extensions in certificates:

- m mandatory – the field **MUST** be present
- x do not use – the field **MUST NOT** be populated
- o optional – the field **MAY** be present

For the criticality of certificate extensions we use the following terminology:

- c critical – the extension is marked critical, receiving applications must be able to process this extension
- nc the extension is marked non-critical, receiving applications that do not understand this extension **MUST** ignore it

For detailed certificate profiles, see the sections below.

6.5.1 CSCA Certificate Profile

The CSCA Certificate profile is defined in [MRTD12].

6.5.2 Arrival Attestation Document Signer Certificate Profile

In Table 2, we list the certificate components of an AAD-Signer certificate.

Table 2: Arrival Attestation Document (AAD) Signer Certificate Profile

Certificate Component	Presence	Remark
Certificate	m	
TBSCertificate	m	see below
signatureAlgorithm	m	dependent on selected algorithm
signatureValue	m	dependent on selected algorithm
TBSCertificate		
version	m	2 (indicating version 3)
serialNumber	m	must be the positive integer that results from interpreting the <i>five character hex-string</i> as a positive integer. This integer uniquely identifies an AAD-Signer Certificate for one CA and subject.

Certificate Component	Presence	Remark
		leading bit must be zero in DER encoding (cf. [MRTD12]).
signature	m	this value MUST be the same as that in the signatureAlgorithm component of certificate sequence
issuer	m	MUST be the value of the subject DN of the CSCA certificate with which the AAD-Signer certificate was signed
validity	m	MUST terminate with Zulu (Z). The seconds element MUST be present. Dates through 2049 must be in UTCTime, represented as YYMMDDHHMMSSZ. Dates in 2050 and beyond MUST be in GeneralizedTime. GeneralizedTime MUST NOT have fractional seconds, and MUST be represented as YYYYMMDDHHMMSSZ. The validity (i.e. difference between notBefore and notAfter) MUST be according to Section 6.4.3
subject	m	the following two MUST be present; other attributes MUST NOT be present. commonName: MUST consist of two uppercase characters in printableString format, that uniquely define the AAD-Signer within the issuing country, and must be present in the header of a AAD-Signer as the Signer reference countryName: MUST consist of the ISO two-letter country code of the AAD-Signer in uppercase characters, printableString format
subjectPublicKeyInfo	m	must adhere to [X509] and [MRTD12]
issuerUniqueID	x	
subjectUniqueID	x	
extensions	m	cf. Table 3 on which extensions should be present. Default values for extensions MUST NOT be encoded.

Extensions are depicted in Table 3. No other certificate extensions must be present.

Table 3: Arrival Attestation Document (AAD) Signer Certificate Profile: Extensions

Extension Name	Presence	Criticality	Remark
AuthorityKeyIdentifier	m	nc	
keyIdentifier	m		
authorityCertIssuer	o		

Extension Name	Presence	Criticality	Remark
authorityCertSerialNumber	o		
ExtKeyUsage	m	c	the EKU extension for each AAD-Signer MUST be populated as: AAD-Signer Certificate, OID 0.4.0.127.0.7.3.5.1.1

6.5.3 Certificate Publication

As the AAD-Signer Certificates are not contained in the digital seal itself, the CA MUST publish its certificates. Publication MUST adhere to the following principles:

1. As soon as a new certificate is created, it MUST be published within a delay of no more than 48 hours.
2. The certificates MUST remain published until their expiration.

6.5.4 Certificate Generation

The certificate generation process consists of the following steps:

- The signature key pair of the AAD-Signer is generated in a cryptographic module or a secure signature creation device.
- A certificate request is created by the AAD-Signer. This request contains the public key (to be certified) of the AAD-Signer which is signed with its private key.
- This certificate request is sent to the CSCA on a secure channel.
- The CSCA verifies the signature of the certificate request, and creates a certificate corresponding to the public key of the AAD-Signer.
- The CSCA returns the certificate to the AAD-Signer.
- The CSCA publishes the certificate.

6.5.5 Certificate Renewal

A new certificate MUST always contain a newly generated key pair. New certificates are regularly created when the corresponding signature keys reach the end of their signing validity period.

6.5.6 Certificate Revocation

An AAD-Signer certificate MUST be revoked in case of a security incident concerning the signature key. The certificate revocation of an AAD-Signer certificate is decided by the country that issues the AAD. Since the AAD-Signer certificate is issued by a CSCA, the revocation of a certificate is published in the CRL of that CSCA, cf. [MRTD12].

6.6 Arrival Attestation Document Validation

Arrival Attestation Documents with digital seals are validated by applying a validation policy. The next section specifies validation criteria and algorithms to generate a validation status in detail.

Figure 5 illustrates the functional architecture for validation. The system relies on validation software which can be deployed on any computer used by the relevant authorities.

The validation software is connected with a reader that takes an image of the document to retrieve the bar code. Optionally (depending on the capabilities of the reader) the printed MRZ of the document is retrieved by using optical character recognition (OCR).

To verify the validity of the signature of the **digital seal**, the validation software **SHOULD BE** synchronized with the PKI publication point at least every 24 hours to retrieve the latest CRLs.

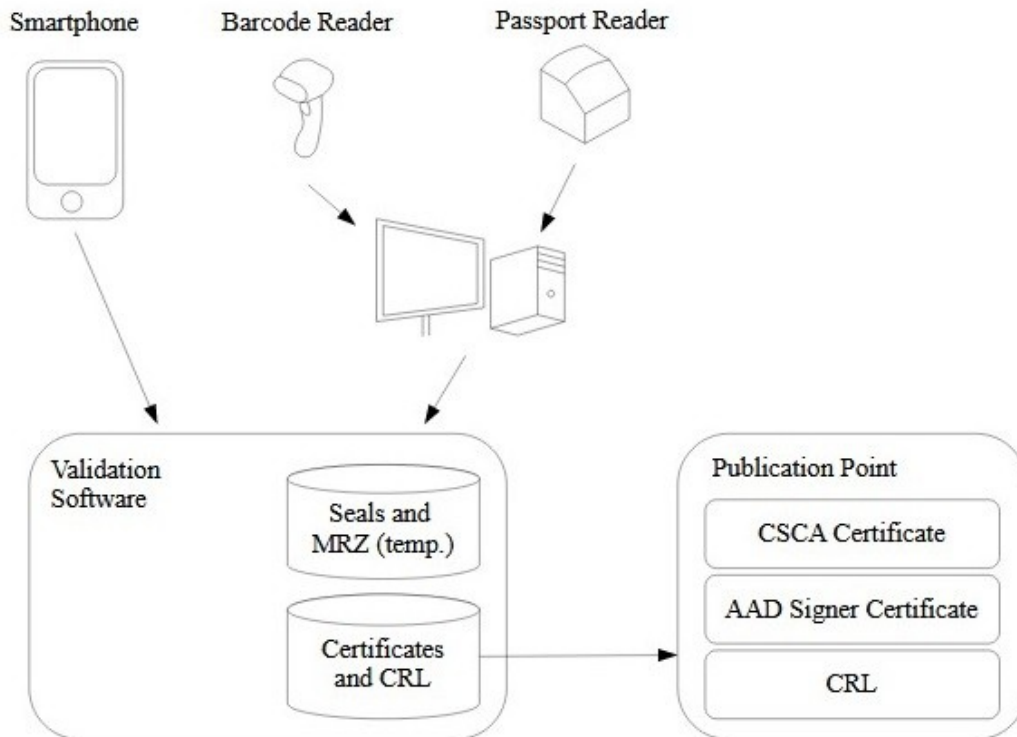


Figure 5: Arrival Attestation Document (AAD) validation

The validation software decodes the digital seal, and optionally OCRs the printed MRZ of the document, validates the signature of the **digital seal**, and applies a validation policy to generate a validation status of the document.

The following data are processed by the validation software:

- Input data provided by readers, i.e. the image of the document, optionally the printed MRZ of the document
- Certificates and CRLs.

6.7 Validation Policy (Informative)

6.7.1 Policy Rules

The Validation Policy is a set of validation rules that allow to determine the validity of the seal on the document. The set of rules how a seal should be verified is described in Appendix D of [DOC 9303-13].

7 Social Insurance Card Profile

This section specifies a profile for digital seals of a Social Insurance Card (SIC). A SIC is a physical document which proves the registration and membership of a person within a national social insurance. The digital seal on a SIC, in addition to human-readable personal data printed on the SIC, guarantees the authenticity and integrity of the document's data and protects against forgery. To verify the tie between a document and its holder, an ID document is required.

7.1 Bar Code Symbology

The seal **MUST** be encoded as a DataMatrix [DTMX] bar code. The symbol size depends on the number of bytes, which are encoded in the bar code. The minimum symbol size of the DataMatrix bar code according to table 7 of [DTMX] **SHOULD** be chosen in a way that the seal fits in the bar code. Given the specified content encoding rules, the maximum required symbol size is 80 x 80 with a maximum raw capacity of 453 bytes (without error-correction).

A worked example is shown in Annex D: Worked Example of a Social Insurance Card.

7.2 Content and Encoding Rules

7.2.1 Header

The Version of the SIC is 0x02.

The *Document Feature Definition Reference* for this use case is 0xFC

The *Document Type Category* for Social Insurance Card is 0x04.

Otherwise, the content of the header is according to [DOC 9303-13].

7.2.2 Document Features of a Seal for a National Social Insurance Card

This section describes the data stored in the digital seal of the Social Insurance Card. These data are:

- Social insurance number (required)
- Surname and first name (required)
- Name at birth (optional)

Other additional document features are not permitted for encoding of the digital seal for the SIC.

7.2.3 Encoding Rules for Document Features

In the following, the digital encoding of document features of a SIC seal is defined.

Social Insurance Number

Tag: 0x01

Min. Length: 8 bytes

Max. length: 8 bytes

Value Type: Alphanumeric

Presence: REQUIRED

Content: A unique alphanumeric string of 12 characters, C40-encoded.

Surname

Tag: 0x02

Min. Length: 1 byte

Max. length: 90 bytes

Value Type: Unicode utf-8

Presence: REQUIRED

Content: A string of max. 30 characters, encoded by UTF-8 as a binary sequence.

First name

Tag: 0x03

Min. Length: 1 byte

Max. length: 90 bytes

Value Type: Unicode utf-8

Presence: REQUIRED

Content: A string of max. 30 characters, encoded by UTF-8 as a binary sequence.

Name at birth

Tag: 0x04

Min. Length: 1 byte

Max. length: 90 bytes

Value Type: Unicode utf-8

Presence: CONDITIONAL

Content: A string of max. 30 characters, encoded by UTF-8 as a binary sequence.

This field is REQUIRED if the birth name differs from the surname. Otherwise, this field MUST be omitted.

7.2.4 Signature

Hashing and signature algorithm defined in [KRYPT] MUST be used for the digital seal.

8 Residence Permit Profile

This chapter specifies the profile for a digital seal on a paper based residence permit (RP) document. This RP document is issued to foreigners allowed to stay in Germany for a temporary time period. It contains the digital seal in addition to the personal data to ensure the integrity and authenticity of the data printed on the document.

8.1 Bar Code Symbolology

The digital seal **MUST** be encoded as a DataMatrix [DTMX] bar code, with 44 x 44 modules (i.e. the maximum raw data capacity is 142 bytes (see [DTMX])). The bar code **MUST** be printed with 600dpi with 8 dots per module side length (i.e. 64 dots per module). All other requirements are according to section 3.1. A worked example of a digital seal for the Residence Permit Document is described in Annex E: Worked Example of a Digital Seal for Residence Permit Documents.

8.2 Content and Encoding Rules

8.2.1 Header

The Version (at start position 0x01) in the header **MUST** be 4, i.e. 0x03 according to [DOC 9303-13].

The *Document Feature Definition Reference* for the Residence Permit Document according to [DOC 9303-13] **MUST** be set to 0xFB.

The *Document Type Category* for the Residence Permit Documents is 0x06. For all other parameters of the header, [DOC 9303-13] applies.

8.2.2 Document Feature of the Digital Seal for the Residence Permit Document

The document features listed below **MUST** be encoded in the bar code for the Residence Permit Document. Other document features **MUST NOT** be present.

Machine Readable Zone (REQUIRED)

Basic Information is encoded using a Machine Readable Zone (MRZ) of a TD2-Format MROTD [MRTD6]. This MRZ contains the following information:

- document code
- issuing state
- surname and first name of the document holder
- document number
- nationality of the document holder
- date of birth of the document holder
- sex of the document holder
- date of expiry

Passport Number (REQUIRED)

This is a String composed of 9 alphanumeric characters.

8.2.3 Encoding Rules for Document Features of the Residence Permit Document

The Document Features encoding for digital seals of the Residence Permit Document is defined by the following data.

MRZ (TD2-Type [MRTD6])

Tag: 0x02

Min. Length: 48 bytes

Max. Length: 48 bytes

Value Type: Alphanumeric

Presence: REQUIRED

Content: The first line and second line of the MRZ of a TD2-MROTD (2*36 chars.) MUST be concatenated and encoded by C40. The filler symbol < in the MRZ is replaced by <SPACE> prior to C40 encoding.

Passport Number

Tag: 0x03

Min. Length: 6 bytes

Max. Length: 6 bytes

Value Type: Alphanumeric

Presence: REQUIRED

Content: The content of the document number is C40-encoded.

8.2.4 Signature

For generation of digital signatures the hashing and signature algorithms according to [KRYPT] SHALL be used.

8.3 Digital Seal Encoding for the Supplementary Sheet (Optional)

The Supplementary Sheet of the Residence Permit Document contains additional information of the document holder, like the domicile and the work permission of the document holder. The issuance of a Supplementary Sheet is optional in general, but if it is issued, the following data SHALL be encoded in the digital seal:

- The MRZ of the Residence Permit Document
- Number of the supplementary sheet

For encoding the digital seal on the supplementary sheet, the same bar code symbology as described in section 8.1 applies.

8.4 Content and Encoding Rules

8.4.1 Header

The Version of the header MUST be 0x03 (at Position 0x01) indicating version number 4 according to [DOC 9303-13].

The *Document Feature Definition Reference* for supplementary sheet according to section 3.2.1 MUST be set to 0xFA.

The *Document Type Category* for supplementary sheet is 0x06.

Otherwise, the content of the header is according to [DOC 9303-13].

Note: The distinction between the Residence Permit and supplementary sheet MUST be done by the Document Feature Definition Reference, because both have the same Document Type Category.

8.4.2 Encoding Rules for Document Features of the Supplementary Sheet

The digital seal of the supplementary sheet SHALL encode the following document features.

MRZ of the Residence Permit Document (TD2-Type [MRTD6])

Tag: 0x04

Min. Length: 48 bytes

Max. Length: 48 bytes

Value Type: Alphanumeric

Presence: REQUIRED

Content: The first line and second line of the MRZ of a TD2-MROTD (2*36 chars.) MUST be concatenated and encoded by C-40. The filler symbol < in the MRZ is replaced by <SPACE> prior to C40 encoding.

Number of supplementary sheet

Tag: 0x05

Min. Length: 6 bytes

Max. Length: 6 bytes

Value Type: Alphanumeric

Presence: REQUIRED

Content: The content of the supplementary sheet number is C40-encoded.

8.5 Public Key Infrastructure

For verifying the signature of the digital seal for Residence Permit Documents, a residence permit (RP)-Signer certificate issued by the CSCA is required. Thus the RP-Signer certificates are issued in a way that allows verification by the CSCA certificate (defined in [MRTD12]). This means the RP-Signer certificate is directly issued by the CSCA without having a SubCA in between. The RP-Signer uses its private key for signing the data stored in the digital seal. The corresponding public key is part of the RP-Signer certificate, which is used to verify the signature on the digital seal.

8.5.1 Certificates

CSCA certificate

The structure of the CSCA certificate is defined in [MRTD12].

Residence Permit Signer Certificate

The encoding of the Residence Permit Signer Certificate is described in Table 4 below.

Table 4: Encoding of the Residence Permit Signer Certificate

Certificate Component	Presence	Remark
Certificate	m	
TBSCertificate	m	see below
signatureAlgorithm	m	dependent on selected algorithm
signatureValue	m	dependent on selected algorithm
TBSCertificate		
version	m	2 (indicating version 3)
serialNumber	m	MUST be the positive integer that results from interpreting the up to <i>three character hex-string</i> as a positive integer. This integer uniquely identifies an RP-Signer Certificate for one CA and subject. The leading bit must be zero in DER encoding (cf. [MRTD12]).
signature	m	This value MUST be the same as that in the signatureAlgorithm component of certificate sequence
issuer	m	MUST be the value of the subject DN of the CSCA certificate with which the RP-Signer certificate was signed
validity	m	MUST terminate with Zulu (Z). The seconds element MUST be present. Dates through 2049 MUST be in UTCTime, represented as YYMMDDHHMMSSZ. Dates in 2050 and beyond MUST be in GeneralizedTime. GeneralizedTime MUST not have fractional seconds, and MUST be represented as YYYYMMDDHHMMSSZ. The validity (i.e. difference between notBefore and notAfter) MUST be according to Section 6.4.3
subject	m	the following two MUST be present; other attributes MUST NOT be present. commonName: MUST consist of <i>two uppercase characters</i> in printableString format, that uniquely define the RP-Signer within the issuing country, and MUST be present in the header of a RP-Signer as the <i>Signer reference</i>

Certificate Component	Presence	Remark
		countryName: MUST consist of the ISO two-letter country code of the RP-Signer in uppercase characters, printableString format
subjectPublicKeyInfo	m	MUST adhere to [X509] and [MRTD12]
issuerUniqueID	x	
subjectUniqueID	x	
extensions	m	All mandatory extensions according to [MRTD12] SHALL be used. In addition, the DocumentType extension SHALL also be present.

8.5.2 Certificate Validity Periods

The same certificate validity periods for the CSCA and Residence Permit Signer as described in section 6.4.3 SHALL be used.

8.5.3 Certificate Publication

The RP-Signer certificates are published as described in [MRTD12].

8.5.4 Certificate Revocation

In case of a security incident (e.g. the private Key is compromised), the RP-Signer certificate MUST be revoked. For revocation the requirements according to [MRTD12] SHALL apply.

9 Address Sticker Profile for the German Identity Card and Residence Permit Card

This chapter specifies the profile for the digital seal of the address sticker for the German Identity Card and the Residence Permit Card. Address stickers are issued to persons that use online means for the compulsory report of a change of address to the municipal office. Address stickers are affixed to the German Identity Card or Residence Permit Card and replace the address printed on the Identity Card or Residence Permit Card. To logically secure the sticker against falsification, parts of the data that are contained on the sticker are protected by the digital seal.

9.1 Bar Code Symbology

The digital seal **MUST** be encoded as DataMatrix [DTMX] bar code, with 40 x 40 modules (i.e. the maximum raw capacity is 112 bytes according to [DTMX]). The bar code **MUST** be printed on a surface of 13,04mm x 13,04mm (quiet zone included) with 7 dots per module side length (i.e 49 dots per module) with 600dpi. All other requirements of section 3.1 apply.

9.2 Content and Encoding Rules

9.2.1 Header

The version at start position (0x01) of the header **MUST** be 0x03 (version 4) according to [DOC 9303-13].

The Document Feature Definition Reference defining the number and encoding of document features **MUST** be 0xF9 according to [DOC 9303-13].

The Document Type Category is defined as follows:

- 0x08 for the address sticker for German Identity Card
- 0x0C for the address sticker for Residence Permit Card

All other parameter of the header are according to section 2.2 of [DOC 9303-13].

9.2.2 Document Features of a Digital Seal for the Address Sticker for the German Identity Card

The following list of document features **MUST** be used for encoding the data in the digital seal.

Document Number (REQUIRED)

A string of 9 alphanumeric characters that identifies the ID document the sticker is attached to.

Official Municipality Code Number (REQUIRED)

Identifies the responsible authority issuing the address sticker containing the new residential address of the applicant as described in [TR-03127]. The country code and leading zeros **SHALL NOT** be encoded. As a result, it is composed of 8 numeric characters².

2 As defined by the Federal Statistical Office
<https://www.destatis.de/DE/Themen/Laender-Regionen/Regionales/Gemeindeverzeichnis/Glossar/amtlicher-gemeindeschluessel.html>

Residential Address (REQUIRED)

A string of maximum 26 C40-encoded characters (see Annex B: C40-Encoding of Strings (Normative)) that is composed of the concatenation of postal code (only for an address in Germany), name of the street and the house number. Only residential addresses in Germany SHALL be encoded in the digital seal. The postal code is composed of exactly 5 numeric characters followed by the name of the street and the house number (21 C40-encoded characters in total). Due to the limited storage capacity of the 40 x 40 bar code, the residential address MUST be converted into a reduced form following the shortening rules as described in [BMI-Letter]. Additional features MUST NOT be used.

9.2.3 Encoding Rules of Document Features for the Address Sticker of German Identity Card

The document features for the address sticker of German identity cards MUST be encoded as described in the following.

Document Number

Tag:	0x01
Min. Length:	6 bytes
Max. Length:	6 bytes
Value Type:	Alphanumeric
Presence:	REQUIRED
Content:	The document number: An alphanumeric string of 9 characters, C40-encoded.

Official Municipality Code Number

Tag:	0x02
Min. Length:	6 Bytes
Max. Length:	6 Bytes
Value Type:	Numeric
Presence:	REQUIRED
Content:	The Official Municipality Code Number: An numeric string of 8 characters, encoded by C40.

Residential Address

Tag:	0x03
Min. Length:	6 bytes
Max. Length:	18 bytes
Value Type:	Alphanumeric
Presence:	REQUIRED
Content:	The residential address is an alphanumeric string of at most 26 characters, C40-encoded. The residential address MUST be encoded by the concatenation of postal code (5 numeric characters), the name of the street and house number following the rules as defined in [BMI-Letter].

9.2.4 Signature

For generating the digital signature, hashing and signature algorithms according to [KRYPT] SHALL be used.

9.3 Public Key Infrastructure

For verifying the digital seal of the address sticker, an address sticker (AS)-Signer certificate is required. These are directly issued by the CSCA, i.e. there is no SubCA in between, and the CSCA-certificate is used to verify the signature of the AS-Signer certificate. The private key of the AS-Signer certificate is used to generate the signature. The public key of the AS-Signer certificate is used to verify the signature of the digital seal.

9.3.1 Certificates

CSCA Certificate

The structure of the CSCA certificate is defined in [MRTD12].

Address Sticker Signer Certificate

The address sticker signer certificate MUST be structured as defined in Table 5 below.

Table 5 Encoding of address sticker signer certificate

Certificate Component	Presence	Remark
Certificate	m	
TBSCertificate	m	see below
signatureAlgorithm	m	dependent on selected algorithm
signatureValue	m	dependent on selected algorithm
TBSCertificate		
version	m	2 (indicating version 3)
serialNumber	m	MUST be the positive integer that results from interpreting the up to <i>three character hex-string</i> as a positive integer. This integer uniquely identifies an AS-Signer Certificate for one CA and subject. The leading bit must be zero in DER encoding (cf. [MRTD12]).
signature	m	this value must be the same as in the signatureAlgorithm component of the certificate sequence
issuer	m	MUST be the value of the subject DN of the CSCA-certificate used to sign the AS-Signer certificate
validity	m	MUST terminate with Zulu (Z). The seconds element MUST be present. Dates through 2049 MUST be in UTCTime, represented as YYMMDDHHMMSSZ. Dates in 2050 and beyond MUST be in GeneralizedTime. GeneralizedTime MUST not have fractional seconds, and MUST be

Certificate Component	Presence	Remark
subject	m	represented as YYYYMMDDHHMMSSZ. The validity (i.e. difference between notBefore and notAfter) MUST be according to section 9.3.2
		the following two MUST be present; other attributes MUST NOT be present.
		commonName: MUST consist of two uppercase characters in printableString format that uniquely define the AS-Signer within the issuing country, and MUST be present in the header of a AS-Signer as the <i>Signer reference</i>
		countryName: MUST consist of the ISO two-letter country code of the AS-Signer in uppercase characters, printableString format
subjectPublicKeyInfo	m	MUST adhere to [X509] and [MRTD12]
issuerUniqueID	x	
subjectUniqueID	x	
extensions	m	all mandatory extensions according to [MRTD12] SHALL be used. In addition, the DocumentType extension SHALL be set to "ST".

9.3.2 Certificate Validity Periods

The validity periods are defined as follows:

CSCA Certificate

The same private key usage period and certificate validity period as defined in section 6.4.3 SHALL be used.

AS-Signer Certificate

Private Key Usage Time: 1 year + 1 month (the 1 month is to allow for smooth roll-over)

Certificate validity period³: Private Key Usage Time + maximum validity period of the German ID Documents.

9.3.3 Certificate Publication

The publication of the AS-Signer certificate MUST be done according to section 6.5.3.

9.3.4 Certificate Generation

The generation of the AS-Signer certificate SHALL follow the steps as described in section 6.5.4.

³ This mean the AS-Signer Certificate has a validity period of 11 years and 1 month (for roll-over), as of August 2021

9.3.5 Certificate Revocation

In case of a security incident concerning the signature key, the AS-Signer certificate SHALL be revoked. The requirements of [MRTD12] SHALL apply.

10 Place of Residence Sticker Profile for German Passports

This chapter specifies the digital seal profile for the place of residence sticker for German passports. These stickers are issued to persons using online notifications for reporting a change of residence. All persons who are in possession of a passport in addition to the ID card will get a sticker containing a digital seal, which is to be placed inside the passport. This sticker contains the document number of the document the sticker is attached to, the place of residence of the passport holder and the issuing authority. The digital seal secures the data printed on the sticker against falsification.

10.1 Bar Code Symbology

The same bar code symbology is used as defined in section 9.1.

10.2 Content and Encoding Rules

10.2.1 Header

The Document Feature Definition Reference MUST be `0xF8` and the Document Type Category `0x0A` according to [DOC 9303-13]. All other parameter MUST be according to section 9.2.1.

10.2.2 Document Features for the digital seal for place of residence stickers

The following list of document features MUST be used for encoding the data in the digital seal. Other features as described below MUST NOT be used.

Document Number (REQUIRED)

A string of 9 alphanumeric characters that identifies the passport the sticker is attached to.

Official Municipality Code Number (REQUIRED)

Identifies the responsible authority issuing the address sticker containing the new residential address of the applicant as described in [TR-03127]. The country code and leading zeros SHALL NOT be encoded. As a result, it is composed of 8 numeric characters.

Postal Code (REQUIRED)

Is a string of 5 numeric characters identifying the place of residence of a person.

10.2.3 Encoding Rules for the digital seal for the place of residence sticker

Document Number

Tag:	0x01
Min. Length:	6 bytes
Max. Length:	6 bytes
Value Type:	Alphanumeric
Presence:	REQUIRED

Content: The document number: An alphanumeric string of 9 characters, C40-encoded.

Official Municipal Code Number

Tag: 0x02

Min. Length: 6 bytes

Max. Length: 6 bytes

Value Type: Numeric

Presence: REQUIRED

Content: The Official Municipal Code Number: A numeric string of 8 characters, C40-encoded.

Postal Code

Tag: 0x03

Min. Length: 4 bytes

Max. Length: 4 bytes

Value Type: Numeric

Presence: REQUIRED

Content: The Postal Code: A numeric string of 5 characters, C40-encoded.

10.2.4 Signature

The same requirements as defined in section 9.2.4 SHALL be applied.

10.3 Public Key Infrastructure

The same requirements as defined in section 9.3 SHALL be applied for the public key infrastructure. This also implies the use of the same bar code signer certificate as described in 9.3.1.

11 Generic Profile for Digital Seals based on [ICAO TR-IDB]

This chapter describes a generic profile for digital seals, that are issued compliant with [ICAO TR-IDB] defining a framework for encoding 2D bar codes as digital seals. Unless stated otherwise, the provisions herein are compliant with the [ICAO TR-IDB]. All allowed message types that can be encoded in the bar code are described in section 11.2.2. The mapping of the message types and the PKI requirements that apply to the specific document profiles are described in the corresponding sections 12 to 17.

11.1 Bar Code Symbology

All digital seals issued according to this technical guideline are compliant with [ICAO TR-IDB] and **MUST** be encoded as DataMatrix [DTMX] bar code. The number of modules and the module size of the DataMatrix bar code depends on the input data defined in the corresponding document profiles. All DataMatrix bar codes **MUST NOT** exceed a printing size of 47mm x 47mm by using a resolution of 600dpi. All other requirements are according to section 3.1.

11.2 Content and Encoding Rules for the Generic Profile

11.2.1 Header

The Country Identifier (2 bytes) **MUST** be the three-letter country code as specified in [Doc 9303-3]. All document profiles **MUST** contain “D<<” as three-letter Country Code and the Country Code **MUST** be encoded as defined in [ICAO TR-IDB] (“<<” **MUST** be replaced by <space> prior the C40 encoding). The signature algorithm used to sign the data (header and message zone, excluding the tag marking the signer certificate zone and everything after it) **MUST** comply with section 3.2.2 as described in [ICAO TR-IDB] and [KRYPT]. All document profiles **MUST** contain the certificate reference and signature creation date as defined in [ICAO TR-IDB], section 3.2. Thus, the header has a constant length of 12 bytes according to [ICAO TR-IDB].

11.2.2 Message Zone

The message zone is encoded according to section 3.3 of [ICAO TR-IDB]. The current section defines all message types following [ICAO TR-IDB] that **MAY** be encoded in the digital seal. Depending on the document profiles defined, the message types are mandatory. Message types that **MAY** be stored in the message zone are a Face Image, MRZ-TD1, MRZ-TD2, MRZ-TD3, the Document Number, Document Reference Number, AZR Number, National Document Type Identifier and the Expiration Date (see [ICAO TR-IDB], section 2.1). Thus, a set of message types are mapped to specific documents. Other message types are not allowed, but may be added in the future of this technical guideline. The message zone **MUST** at least contain one message type for each document profile. The general encoding of the aforementioned message type is described in the following:

MRZ (TD1-Type [MRTD5])

Tag: 0x07

Min. Length: 60 bytes

Max. Length: 60 bytes

Value Type: Alphanumeric
Presence: OPTIONAL
Content: The encoding of MRZ (TD1) SHALL be as defined in [ICAO TR-IDB]. The filler symbol < in the MRZ MUST be replaced by <SPACE> prior to C40-encoding.

MRZ (TD3-Type [MRTD4])

Tag: 0x08
Min. Length: 60 bytes
Max. Length: 60 bytes
Value Type: Alphanumeric
Presence: OPTIONAL
Content: The encoding of MRZ (TD3) SHALL be as defined in [ICAO TR-IDB]. The filler symbol < in the MRZ MUST be replaced by <SPACE> prior to C40-encoding.

Face Image

Tag: 0x80
Min. Length: 900 bytes
Max. Length: 1000 bytes
Value Type: Byte
Presence: OPTIONAL
Content: The input image MUST be a JPEG2000 lossless compressed image with an image size of either 600 x 800 or 622 x 800 pixels (according to [BSI-TR-03121-3]). These images SHALL be compressed to maximum of 1000 bytes. This JPEG2000 input image MUST first be converted to an 8-bit gray scale image. After that, the images are scaled to a final image size of 100 x 129 pixels for the input image of 622 x 800 pixels and 96 x 128 pixels for input image of 600 x 800 pixels and then compressed with maximum of 1000 bytes image size (i.e. the image is encoded in compressed JPEG2000 format). The image with the best quality and less deviations from 1000 bytes should be used. After that, the compressed JPEG2000 image with a maximum size of 1000 bytes is stored in the bar code.

MRZ (TD2-Type [MRTD6])

Tag: 0x81
Min. Length: 48 bytes
Max. Length: 48 bytes
Value Type: Alphanumeric
Presence: OPTIONAL
Content: The first line and second line of the MRZ of a TD2-MROTD (2*36 chars.) MUST be concatenated and encoded by C40. The filler symbol < in the MRZ MUST be replaced by <SPACE> prior to C40-encoding.

Document Reference Number

Tag: 0x82
Min. Length: 6 bytes
Max. Length: 6 bytes

Value Type: Alphanumeric

Presence: OPTIONAL

Content: The content of the Document Reference Number is C40-encoded. It encodes the document number of the reference document, i.e. the document used for identity verification.

AZR Number

Tag: 0×83

Min. Length: 8 bytes

Max. Length: 8 bytes

Value Type: Numeric

Presence: OPTIONAL

Content: The AZR number is a numeric string of 12 characters. The content of the AZR number is C40-encoded.

Expiration Date

Tag: 0×84

Min. Length: 4 bytes

Max. Length: 4 bytes

Value Type: Date

Presence: OPTIONAL

Content: The date is encoded by four bytes, where the first byte encodes a date mask to differentiate between known and unknown parts of the date. The remaining three bytes are encoded by converting the date into a positive integer by concatenating months, days and years as defined in [ICAO TR-IDB].

Document Number

Tag: 0×85

Min. Length: 6 bytes

Max. Length: 6 bytes

Value Type: Alphanumeric

Presence: OPTIONAL

Content: The content of the document number is C40-encoded. This encodes the document number of the document the digital seal is belonging to.

National Document Type Identifier

Tag: 0×86

Min. Length: 1 byte

Max. Length: 1 byte

Value Type: Byte

Presence: OPTIONAL

Content: The National Document Type Identifier is encoded in one byte. It clearly identifies the corresponding document the data in the digital seal belongs to. The corresponding value is defined in the

specific document profiles. The National Document Type Identifier is independent from the document type of the MRZ.

11.2.3 Signer Certificate Zone

The signer certificate zone is absent for all profiles according to this technical guideline. The signer certificate **MUST** be stored outside the digital seal on the bar code decoder side (i.e. on the side the digital signature of the seal is verified).

11.2.4 Signature Zone

All digital seals **MUST** be signed. For signature generation, the algorithms and key lengths according to [KRYPT] **MUST** be used. In addition, all requirements as defined in section 3.5 in [ICAO TR-IDB] **MUST** be used.

11.3 Public Key Infrastructure

For verifying the signature of the digital seal, an IDB signer certificate is required. This certificate is directly issued by the CSCA (i.e. there is no SubCA in between) and the CSCA certificate is used to verify the signature of the IDB signer certificate. The corresponding private key is used for signature generation.

11.3.1 Certificates

CSCA certificate

The CSCA certificate **MUST** be according to [MRTD12].

Bar Code Signer certificates for IDB:

The bar code signer certificate for all IDB **MUST** be constructed as defined in the following Table 6.

Table 6: Certificate structure for IDB signer certificates according to [ICAO TR-IDB]

Certificate Component	Presence	Remark
Certificate	m	
TBSCertificate	m	see below
signatureAlgorithm	m	dependent on selected algorithm
signatureValue	m	dependent on selected algorithm
TBSCertificate		
version	m	2 (indicating version 3)
serialNumber	m	MUST be a positive integer and maximum 20 Octets.
signature	m	this value must be the same as in the signatureAlgorithm component of the certificate sequence
issuer	m	MUST be the value of the subject DN of the CSCA certificate used to sign the IDB signer certificate

Certificate Component	Presence	Remark
validity	m	MUST terminate with Zulu (Z). The seconds element MUST be present. Dates through 2049 MUST be in UTCTime, represented as YYMMDDHHMMSSZ. Dates in 2050 and beyond MUST be in GeneralizedTime. GeneralizedTime MUST not have fractional seconds, and MUST be represented as YYYYMMDDHHMMSSZ. The validity (i.e. difference between notBefore and notAfter) MUST be according to section 9.3.2
subject	m	the following two MUST be present; other attributes MUST NOT be present. commonName: MUST consist of <i>two uppercase characters</i> in printableString format, that uniquely define the IDB signer within the issuing country. countryName: MUST consist of the ISO two-letter country code of the IDB signer in uppercase characters, printableString format.
subjectPublicKeyInfo	m	MUST adhere to [X509] and [MRTD12]
issuerUniqueID	x	
subjectUniqueID	x	
extensions	m	all mandatory extensions according to [MRTD12] SHALL be used. The extendedKeyUsage as defined in [ICAO TR-IDB] MUST be present. In addition, the DocumentType extension as defined in [MRTD12] SHALL be encoded. The specific list of document types that MUST be used are defined in the corresponding profiles. Note that, the document type of the IDB signer certificate is different from the document types of the MRZ of the corresponding documents. The document type of the IDB signer certificate MUST be according to Table 7.

11.3.2 Certificate Validity Period

The specific validity period for IDB signer certificates depend on the validity period of its corresponding document. For the document profiles mentioned below, the following certificate validity periods MUST be applied:

CSCA Certificate

The same private key usage period and certificate validity period as defined in section 6.4.3 SHALL be used.

IDB Signer Certificates

The IDB signer certificate validity period for the documents defined in section 12 to 17 MUST be as defined in Table 7 below.

Table 7: Validity periods for IDB signer certificates

Documents with DocTypes for IDB Signer Certificates ⁴	Private Key Usage Time	Certificate Validity Period ⁵	Max. document validity period
DA	1 year + 1 month	11 years + 1 month	10 years
DB	1 year + 1 month	5 years + 1 month	4 years
NA	1 year + 1 month	1 year + 4 months	3 months

11.3.3 Certificate Publication

The publication of the certificates SHALL follow the principles as described in section 6.5.3.

11.3.4 Certificate Generation

The generation process for the IDB signer certificates MUST follow the process according section 6.5.4.

11.3.5 Certificate Renewal

The requirements for certificate renewal according to section 6.5.5 SHALL apply.

11.3.6 Certificate Revocation

For certificate revocation, the requirements according to [MRTD12] MUST apply.

4 All documents with document types contained in the corresponding document type extension of IDB signer certificates (see document profiles of section 12 to 17) that are allowed to be signed with the corresponding Document Signer private key.

5 Certificate validity period = Private key usage time + max. document validity period

12 Profile for Temporary or Substitutional Travel and ID Documents based on [ICAO TR-IDB]

This chapter specifies the profile for digital seals for temporary or substitutional travel and ID documents issued in Germany. These temporary or substitutional travel and ID documents are issued to applicants who urgently need a passport or ID document, e.g. in case a document is lost or stolen. The corresponding documents are only allowed to be used for the specific use case it is issued for. The following profile defines the document features to be stored in the digital seal to ensure the integrity and authenticity of the paper-based documents.

12.1 Bar Code Symbology

The digital seals for temporary documents or substitutional travel and ID documents **MUST** be encoded as DataMatrix [DTMX] bar code with 132 x 132 modules (i.e. the maximum raw capacity is 1,301 bytes or 1,954 alphanumeric character). The bar code **MUST** be printed with 600dpi and 8 dots per module side length (i.e. 64 dots per module). A worked example for a digital seal compliant with [ICAO TR-IDB] is described in Annex I: Worked Example for Temporary or Substitutional Travel and ID Documents. All other requirements of section 11.1 **SHALL** apply.

12.2 Content and Encoding Rules

12.2.1 Header

The header **MUST** be encoded as defined in section 11.2.1.

12.2.2 Message Zone

This sections maps the encoding of the message types as described in section 11.2.2 to the different temporary or substitutional travel and ID documents. The mapping is defined in Table 8 below. For documents that are used as emergency travel documents (ETDs), the message type (tag: 0x02) is required as defined in [ICAO TR-IDB].

Table 8: List of message types for temporary or substitutional travel and ID documents

Tags ⁶ of message types	Message Types belonging to documents	Document Types in MRZ	National Document Type Identifier (Tag 0x86)	DocTypes for IDB Signer Certificates
0x80, 0x81, 0x84, 0x86	Substitute Identity ⁷ Document (Ausweisersatz)	AI	0x01	DB
0x02, 0x80, 0x84, 0x86	Emergency Travel Document (Reiseausweis als Passersatz)	PU	0x02	NA

6 All documents have more then one message type tag. Documents that contain the message type tag (0x02) are ETDs according to [ICAO TR-IDB]. In addition, these documents also contain national message type tags that are defined in section 11.2.2.

7 The last check digit of the MRZ-TD2 at index 36 in line two is computed over the issued date (index 30 -35 of MRZ) instead of calculating the check digit according to ICAO.

0x02, 0x80, 0x84, 0x86	Emergency Travel Document (Notreiseausweis)	PU	0x03	NA
0x02, 0x80, 0x84, 0x86	Temporary Identity Card (Vorläufiger Personalausweis)	IT	0x04	NA
0x08, 0x80, 0x84, 0x86	Temporary Official Passport (Vorläufiger Dienstpäss)	PO	0x05	DB
0x08, 0x80, 0x84, 0x86	Temporary Passport (Vorläufiger Reisepass)	PP	0x06	DB
0x08, 0x80, 0x84, 0x86	Temporary Diplomatic Passport (Vorläufiger Diplomatenpass)	PD	0x07	DB
0x08, 0x80, 0x84, 0x86	Temporary Travel Document – Convention of 28 July 1951 (Vorläufiger Reiseausweis für Flüchtlinge)	PR	0x08	DB
0x08, 0x80, 0x84, 0x86	Temporary Travel Document – Convention of 28 September 1954 (Vorläufiger Reiseausweis für Staatenlose)	PS	0x09	DB
0x08, 0x80, 0x84, 0x86	Temporary Travel Document for Foreigners (Vorläufiger Reiseausweis für Ausländer)	PT	0x0A	DB
0x80, 0x81, 0x84, 0x86	Temporary Identity Document (Ersatz- Personalausweis)	IX	0x0F	DB

All message types according to Table 8 MUST contain a Face Image (tag: 0x80), the expiration date of the document (tag: 0x84), its National Document Type Identifier (tag: 0x86) and either MRZ-TD2 or MRZ-TD3 according to section 11.2.2 in its bar code. The value for National Document Type Identifier SHALL be according to Table 8. The documents with message type (0x02) MUST be encoded as ETD according to [ICAO TR-IDB] and contain the MRZ-TD2 directly encoded in the data structure of ETDs. Except from this, the Substitute Identity Document and the Temporary Identity Document SHALL encode the MRZ-TD2 (tag: 0x81) as defined in section 11.2.2. All other documents contain the message type of MRZ-TD3 (tag: 0x08) according to [ICAO TR-IDB].

12.2.3 Signature Zone

The signature zone begins with tag 0x7F and the header and message zone MUST be signed by algorithms as defined in [KRYPT]. In addition, the requirements as defined in section 11.2.4 SHALL apply.

12.3 Public Key Infrastructure

For the public key infrastructure, all requirements from section 11.3 SHALL apply. In addition, the following requirements for temporary documents apply.

12.3.1 Certificate Validity Periods

The certificate validity period MUST be as defined in section 11.3.2.

12.3.2 Certificate Extension

The certificate extensions as defined in Table 6 SHALL be encoded in the IDB signer certificate. For the DocumentType extension, the list of document types for IDB signer certificates according to Table 8 SHALL be encoded in the certificate.

13 Profile for Immigration Legal Documents based on [ICAO TR-IDB]

These documents have different purposes. The "Permission to Retain Pending the Asylum Decision" (DE: Aufenthaltsgestattung zur Durchführung des Asylverfahrens) is issued as a document for the conduct of the asylum procedure. The document "Temporary Suspension of Deportation" (DE: Aussetzung der Abschiebung) is issued to foreign nationals who are obliged to leave the country if no return is possible during the validity of this document. The document "Arrival Attestation" (DE: Ankunftsachweis) is issued to a foreigner who is a third-country national if he has made an asylum application but has not yet submitted an asylum application. This chapter describes the encoding of the data stored in the digital seal, in addition to the data printed on the document, to ensure integrity and authenticity of the document.

13.1 Bar Code Symbology

The digital seals for temporary documents MUST be encoded as DataMatrix [DTMX] bar code with 132 x 132 modules (i.e. the maximum raw capacity is 1,301 bytes or 1,954 alphanumeric character). The bar code MUST be printed with 600dpi and 8 dots per module side length (corresponding to 64 dots per module). All other requirements of section 11.1 SHALL apply.

13.2 Content and Encoding Rules

13.2.1 Header

The header MUST be according to section 11.2.1.

13.2.2 Message Zone

This section maps the message types as described in section 11.2.2 to the different documents listed in Table 9 below.

Table 9: Message types for Immigration Legal Documents

Tag of message types	Message Types belonging to the documents	Document Type in MRZ	National Document Type Identifier (Tag 0x86)	DocTypes for IDB Signer Certificates
0x80, 0x81, 0x83, 0x86	Permission to Retain Pending the Asylum Decision ⁸ (Aufenthaltsgestattung zur Durchführung des Asylverfahrens)	AP	0x0B	DB
0x80, 0x81, 0x83, 0x86	Temporary Suspension of Deportation ⁹ (Aussetzung der Abschiebung)	AA	0x0C	DB
0x80, 0x81, 0x83, 0x86	Arrival Attestation (Ankunftsnaechweis)	AU	0x0D	DB

All message types according to Table 9 MUST contain a Face Image (tag: 0x80), an MRZ-TD2 (tag: 0x81), an AZR Number (tag: 0x83) and the National Document Type Identifier (tag: 0x86) according to section 11.2.2. Other message types SHALL NOT be used.

13.2.3 Signature Zone

The signature MUST follow the requirements as defined in section 11.2.4.

13.3 Public Key Infrastructure

All requirements for the public key infrastructure SHALL apply as defined in section 11.3. Furthermore, the validity period and the Document Type extension are defined below.

13.3.1 Certificate Validity Periods

The certificate validity period MUST be as defined in section 11.3.2.

13.3.2 Certificate Extension

The certificate extensions as defined in Table 6 SHALL be encoded in the IDB signer certificate. For the DocumentType extension, the list of document types for the IDB signer certificates according to Table 9 SHALL be encoded in the certificate.

⁸ See footnote 7

⁹ See footnote 7

14 Profile for Provisional Residence Document based on [ICAO TR-IDB]

The Provisional Residence Document (DE: Fiktionsbescheinigung) is a paper-based document, that is used to prove existence of a temporary right of residence in Germany, which regularly arises when an application for the issuance or extension of a residence permit is submitted to the immigration authority. It contains a digital seal with the data that are printed on the document, to save the document against falsification.

14.1 Bar Code Symbology

The digital seals for the Provisional Residence Document **MUST** be encoded as DataMatrix [DTMX] bar code with 48 x 48 modules (i.e. the maximum raw capacity is 172 bytes or 259 alphanumeric character). The bar code **MUST** be printed with 600dpi and 22 dots per module side length (i.e. 484 dots per module). All other requirements of section 11.1 **SHALL** apply.

14.2 Content and Encoding Rules

14.2.1 Header

The header **MUST** be according to section 11.2.1

14.2.2 Message Zone

The message zone **MUST** be encoded according to section 11.2.2 and maps the message types to the specific document type of the Provisional Residence Document as defined in Table 10.

Table 10: Message types for Provisional Residence Document

Tag of message types	Message Types belonging to the documents	Document Type in MRZ	National Document Type Identifier (Tag 0x86)	DocTypes for IDB Signer Certificates
0x81 ¹⁰ , 0x82, 0x83, 0x86	Provisional Residence Document (Fiktionsbescheinigung)	AB	0x0E	DB

The message types according to Table 10 **MUST** contain a MRZ-TD2 (tag: 0x81), the Document Reference Number (tag: 0x82), an AZR-Number (0x83) and the National Document Type Identifier (tag: 0x86).

The value for the National Document Type Identifier of the provisional residence document **MUST** be 0x0E. Other message types **SHALL NOT** be used. The document type in the MRZ for the Provisional Residence Document is "AB" as defined in Table 10.

14.2.3 Signature Zone

The signature **MUST** be generated as defined in 11.2.4.

¹⁰ See footnote 7

14.3 Public Key Infrastructure

For the public key infrastructure all requirements as defined in section 11.3 SHALL apply. The certificate validity period and the encoded document type for the DocumentType extension (see [MRTD12]) are defined in the following sections.

14.3.1 Certificate Validity Periods

The certificate validity period MUST be as defined in section 11.3.2.

14.3.2 Certificate Extension

The certificate extensions as defined in Table 6 SHALL be encoded in the IDB signer certificate. The document type in the DocumentType extension of the IDB signer certificate for the Provisional Residence Document MUST be encoded as defined in *Table 10*.

15 Document Certifying Permanent Residence based on [ICAO TR-IDB]

The Document Certifying Permanent Residence (DE: Bescheinigung des Daueraufenthalts) is an official confirmation in Germany for citizens of the European Union and the European Economic Area allowing permanent right of residence in Germany. This paper based document will contain a digital seal according to [ICAO TR-IDB].

15.1 Bar Code Symbology

The digital seals for documents certifying permanent residence MUST be encoded as DataMatrix [DTMX] bar code with 48 x 48 modules (i.e. the maximum raw capacity is 172 bytes or 259 alphanumeric character). The bar code MUST be printed with 600dpi and 22 dots per module side length (i.e. 484 dots per module). All other requirements of section 11.1 SHALL apply.

15.2 Content and Encoding Rules

15.2.1 Header

The header MUST be encoded as defined in section 11.2.1.

15.2.2 Message Zone

The message zone MUST be encoded as defined in section 11.2.2 and maps the defined message type to this specific profile as defined in Table 11.

Table 11: Message types for Document Certifying Permanent Residence

Tag of message types	Message Types belonging to the documents	National Document Type Identifier (Tag 0x86)	DocTypes for IDB Signer Certificates
0x82, 0x85, 0x86	Document Certifying Permanent Residence (Bescheinigung des Daueraufenthalts)	0x10	DA

The message type according to Table 11 MUST contain the Document Reference Number (tag: 0x82), the Document Number of Document Certifying Permanent Residence (tag: 0x85) and the National Document Type Identifier (tag: 0x86).

The value for the National Document Type Identifier for the Document Certifying Permanent Residence is 0x10. Other message types SHALL NOT be encoded in the digital seal.

15.2.3 Signature Zone

The requirements for the signature generation MUST be as defined in section 11.2.4.

15.3 Public Key Infrastructure

For the public key infrastructure, all requirements as defined in section 11.3 SHALL apply. The certificate validity period and the encoded document type for the DocumentType extension (see [MRTD12])) are defined the following sections.

15.3.1 Certificate Validity Period

The certificate validity period MUST be as defined in section 11.3.2.

15.3.2 Certificate Extensions

The certificate extensions as defined in Table 6 SHALL be encoded in the IDB signer certificate. For the DocumentType extension, the document type for the IDB signer certificate SHALL be encoded as defined in Table 11.

16 Frontier Worker Permit Profile based on [ICAO TR-IDB]

Frontier Worker Permits (DE: Grenzgängerkarte) are issued to citizens of third level countries (neither an EU member state nor an EEA state) who study or work in Germany, while legally living in a country bordering Germany, and returning to this country of residence at least once a week. Thus, the Frontier Worker Permit allows people to work or study in Germany, without having a permanent residence in Germany. The digital seal printed on the document ensures the integrity and authenticity of the document.

16.1 Bar Code Symbology

The digital seals for temporary documents **MUST** be encoded as DataMatrix [DTMX] bar code with 132 x 132 modules (i.e. the maximum raw capacity is 1,301 bytes or 1,954 alphanumeric character). The bar code **MUST** be printed with 600dpi and 8 dots per module side length (i.e. 64 dots per module). All other requirements of section 11.1 **SHALL** apply.

16.2 Content and Encoding Rules

16.2.1 Header

The header **MUST** be encoded as defined in section 11.2.1.

16.2.2 Message Zone

The message zone **MUST** be encoded as defined in section 11.2.2 and maps the message types to the Frontier Worker Permit document as defined in Table 12 .

Table 12: Message types for Frontier Worker Permits

Tag of message types	Message Types belonging to the documents	Document Type in MRZ	National Document Type Identifier (Tag 0x86)	DocTypes for IDB Signer Certificates
0x80, 0x81 ¹¹ , 0x82, 0x86	Frontier Worker Permits (Grenzgängerkarte)	AG	0x11	DB

The message type as defined in Table 12 **MUST** contain a Face Image (tag: 0x80), the MRZ-TD2 (tag: 0x81), the Document Reference Number (tag: 0x82) and the National Document Type Identifier (tag: 0x86).

The value for the National Document Type Identifier for the Frontier Worker Permit **SHALL** be 0x11. Other message types **SHALL NOT** be used. The document type in the MRZ **MUST** be set to “AG”.

16.2.3 Signature Zone

The signature **SHALL** be encoded as defined in 11.2.4.

¹¹ See footnote 7

16.3 Public Key Infrastructure

For the public key infrastructure, the requirements of section 11.3 SHALL be used. The validity of the IDB signer certificate for the Frontier Worker Permit and the corresponding document type is defined in the following sections.

16.3.1 Certificate Validity Period

The certificate validity period MUST be as defined in section 11.3.2.

16.3.2 Certificate Extensions

The certificate extensions from in section 11.3.1 MUST be encoded in the corresponding IDB signer certificate. For the DocumentType extension according to [MRTD12], the document type for the IDB signer certificate MUST be as defined in Table 12.

17 Supplementary Sheet for Residence Permit Card based on [ICAO TR-IDB]

The Supplementary Sheet to the Residence Permit Card (DE: Zusatzblatt zur Aufenthaltskarte) is a paper-based document that contains additional information from the document holder, e.g. on employment or studies, which are not printed on the corresponding residence permit card body. The supplementary sheet is only valid in combination with the residence permit card.

17.1 Bar Code Symbology

The digital seals for temporary documents **MUST** be encoded as DataMatrix [DTMX] bar code with 64 x 64 modules (i.e. the maximum raw capacity is 277 bytes or 418 alphanumeric character). The bar code **MUST** be printed with 600dpi and 16 dots per module side length (i.e. 256 dots per module). All other requirements of section 11.1 **SHALL** apply.

17.2 Content and Encoding Rules

17.2.1 Header

The header **MUST** be encoded as defined in section 11.2.1.

17.2.2 Message Zone

The message zone **MUST** be encoded as defined in section 11.2.2. The message types for supplementary sheet for residence permit card **MUST** be composed of the following elements as defined in Table 13.

Table 13: Message types for Supplementary Sheet of Residence Permit Card

Tag of message types	Message Types belonging to the documents	Document Type in MRZ	National Document Type Identifier (Tag 0x86)	DocTypes for IDB Signer Certificates
0x07, 0x82, 0x85, 0x86	Supplementary Sheet of Residence Permit Card (Zusatzblatt zur Aufenthaltskarte)	AF, AR or AS	0x12	DA

The message types according to Table 13 **MUST** contain the MRZ-TD1 (tag: 0x07) of Residence Permit Card, the Document Number of Residence Permit Card (tag: 0x82), the Document Number of Supplementary Sheet (tag: 0x85) and the National Document Type Identifier (tag: 0x86).

The value of the National Document Type Identifier for the Supplementary Sheet for the Residence Permit Card **MUST** be 0x12. Other message types **SHALL NOT** be used. The document type in the MRZ **MUST** either be AF, AR or AS as defined in Table 13 depending on the underlying Residence Permit document.

17.2.3 Signature Zone

The signature **SHALL** be encoded as defined in section 11.2.4.

17.3 Public Key Infrastructure

For the public key infrastructure the requirements as defined in section 11.3 SHALL be used. The validity of the IDB signer certificate for the Supplementary Sheet for the Residence Permit Card and the corresponding document type are defined in the following sections.

17.3.1 Certificate Validity Period

The certificate validity period MUST be as defined in section 11.3.2.

17.3.2 Certificate Extensions

The certificate extensions from section 11.3.1 MUST be encoded in the corresponding IDB signer certificate. For the DocumentType extension according to [MRTD12], the document type for the IDB signer certificate MUST be as defined in Table 13.

18 Biometric Data in Digital Seals (Informative)

This chapter gives a general overview of aspects of storing biometric data in a digital seal and how biometric features could be protected with biometric template protection functions. This chapter has only informative character.

18.1 Introduction

Today, biometric data (e.g. fingerprints, iris patterns, face, etc.) play an increasingly important role for user authentication and identification and could be used with or instead of other authentication mechanisms. A major advantage of biometric data is individuality, meaning that every human has individual biometric features. However, biometric matching and securing biometric systems against spoofing and attacks (e.g. presentation or injection attacks) are difficult topics and out of scope for this document. The theoretical use-case for biometric data in digital seals is strictly limited to attended biometric verification with physical presence of both the document holder and an operator.

Storing biometric data in a digital seal would enable biometric verification whether the document holder is indeed the document owner. This functionality is especially relevant in case of identity verification, like the use case “arrival attestation” (see Chapter 6) and travel documents, which could enable detection of the document being used by an unauthorized person. Biometric data in combination with digital seals could be used to bind an object (i.e. document) to a person, which is one of the main goals in government-issued documents.

In addition to biometric data, document data have to be stored in a digital seal to reach the goal of document integrity and authenticity. Only the combination of biometric and document data guarantees the authenticity and integrity of a document and binding to its holder. One major problem when storing biometric data in digital seals is that the storage capability of a bar code is easily exceeded because storing biometric features (i.e. the data itself or biometric templates) requires a comparatively large amount of storage capacity. Today, there is no practical available solution to solve this problem in general, but prototypical implementations are promising. Nevertheless, it has to be stressed that the storage space of a bar code is restricted and for this reason a suitable encoding according to the selected bar code has to be chosen. The following examples are given to show how the size of stored data could be decreased:

1. One could store the document data (or in case of a huge amount of data just a hashvalue) and additionally a link to a database, where the biometric reference data are stored (i.e. biometric data are not directly stored in the seal). For example, document data like an MRZ (Machine-Readable-Zone) and a unique identifier which points to the corresponding database entry could be stored in the digital seal of the Arrival Attestation Document, which is described in Chapter 6.
2. Biometric data could be stored as a biometric template which could be constructed with the help of Biometric Template Protection methods. The advantages of some Template Protection algorithms are that biometric data are stored in a compressed and protected way (and thus use less space than directly storing the biometric data itself) and it is resistant against linkability- and cross-matching attacks.

Storing unprotected biometric data in a digital seal may cause problems of privacy and identity theft. Thus, biometric data must be protected but cannot be hashed with a cryptographic one-way function because of measuring errors, which leads to a different hashvalue with high probability. For this reason, the protected biometric data must provide an error-correction mechanism (e.g. Reed-Solomon-Codes) to correctly verify the authenticity of the biometric feature for different captures of the same biometric feature.

Techniques to generate error tolerant protected biometric reference data without the need for additional encryption and the corresponding key management are referred to as *Biometric Template Protection*. The ISO/IEC International Standard [ISO 24745] provides requirements and guidelines for the secure and privacy-compliant management and processing of biometric information. The following section describes the goal and advantages as well as attacks on biometric template protection functions.

18.2 Biometric Template Protection

Template Protection methods are used to save biometric data in a protected way. This means to ensure unlinkability which guarantees that two biometric templates from the same fingerprint cannot be linked and the protection against retrieval of the biometric feature. In case of travel documents, linkability of biometric templates is not a threat for digital seals because these documents clearly carry the user's real identity. Today, there are several methods that fulfill these requirements and are suitable for use in digital seals.

The goal of applying Biometric Template Protection methods for biometric reference information stored in the digital seal is to deny any unauthorized access and disclosure of the biometric features that were used to generate the reference information. To reach these goals, the biometric feature (e.g. fingerprints) is first recorded and extracted by a scanner. Afterwards, for example the extracted feature's minutiae are used to construct the protected biometric template. Each protected biometric template consists of a secret information called pseudonymous identifier (e.g. a hashvalue over the concatenation of coefficients of a randomly chosen polynomial) and auxiliary data, which are stored in plaintext and are used to reconstruct the secret information with the help of error-correction codes. One of the biggest challenge in using error-correction codes in combination with biometric template protection is the ratio between FAR (False Acceptance Rate) and FRR (False Rejection Rate). FAR describes the probability of unauthorized persons who are wrongly recognized as valid and the FRR describes the probability of authorized person who are wrongly not recognized as such. Error-correction and error rates are strongly associated. If a biometric algorithm has a high error tolerance and can correct many errors (i.e. lower FRR), the FAR is usually high. It is obvious that a high FAR could endanger the biometric authentication process and a high FRR could cause a problem of acceptance of biometric algorithms in general.

In addition to the aforementioned security requirements, biometric templates could be stored in compressed and protected ways instead of storing the fingerprint images, which allows the protected template to be stored in the digital seal.

There are several attacks on biometric applications which can not be prevented with template protection functions. For example, if an attacker has a biometric fake ("Presentation Attack Instrument", PAI) of an authorized person, he could be able to get access to the information system. Presentation Attack Detection (PAD) is a separate problem and is out of scope of this document. Another attack is the FAR attack vector. If attackers have access to a large database of biometric features (e.g. fingerprints), these can be used as queries to find a match between the query template (e.g. fingerprint) and the secure part of the template.

19 Further Use Cases (Informative)

In this section, some examples are given to illustrate possible additional deployments of digital seals with various documents, and possible implementations of verification methods. These deployments are only possibilities and are not actual use-cases but they could give a hint in which applications, digital seals also could be used.

19.1 Document Example 1 – Temporary Passport

In the example of a temporary passport, the digital seal protects the following elements of the passport.

- The **MRZ**, which is optically read and decoded using OCR. This feature can be either verified explicitly, by storing a hash value of the contents as a feature string in the digital seal, or implicitly. In the latter case, the MRZ has to be read by (an error-tolerant) OCR technique and hashed prior to verifying the signature.
- The **height of the document holder** printed on the second page of the passport. This text can be optically read and decoded using OCR. Since this text can be represented by only three characters, explicit verification should be used by storing the text as feature string in the digital seal and not a hash value.
- The **facial image** printed in the passport, which is optically read. The scanned image and the corresponding verification data decoded from the digital seal are processed by an error-tolerant facial image verification method.
- An **OVD (Optically variable device)** in the passport. The OVD type is specified by a corresponding data block in the digital seal. No further verification data needs to be stored. During verification of the digital seal, this identifier is decoded from the seal, and can be mapped to a human-comprehensible description of the OVD and displayed to an operator to allow verification of the OVD by other means.
- The digital seal is encoded in a 48×48 DataMatrix bar code having 172 bytes information capacity (bar code's error correction bits excluded). For each of the above document features, a data block containing the corresponding feature verification data is stored in the seal.

19.2 Examples for Feature Processing Methods

19.2.1 MRZ Verification

It can sometimes be challenging to store all data printed on the document within the bar code. An alternative is to just store the signed hash of the data within the bar code, and recreate the data by error-tolerant methods (e.g. error-tolerant OCR) for verification. This is called *implicit verification*. An implicit verification method for a MRZ is given as an example here. It is based on syndrome decoding of Reed-Solomon-Codes.

Each character of the MRZ is uniquely encoded as an element of a finite field, whose size q must be larger than the set of possible characters (37 in the case of an MRZ) to allow unique encoding, but must also be larger than the length of the MRZ to allow Reed-Solomon-Decoding of the MRZ over this field.

This results in a representation of the MRZ as fixed-length vector over the finite field. For error correcting coding, a (q, k, n) -Reed-Solomon code is chosen, so that the length n of the code words equals the length of the MRZ and is smaller than the field size q . The redundancy $(n - k)$ is twice the number of characters that should be correctable.

During generation of the digital seal, the MRZ is encoded as a vector $\mathbf{m}$ over the finite field and the syndrome of this vector is computed for the selected Reed-Solomon-Code. The binary encoding of the syndrome is thus stored as verification data. Furthermore, the feature string is defined as the MRZ content.

Both the feature string (MRZ content string) and the feature verification data are concatenated and used as the MRZ's contribution to the signature input.

Feature String = MRZ content

Verification Data = syndrome of MRZ content

Signature input of feature = MRZ content | syndrome

During verification, the following procedure is applied to recover the vector $\mathbf{m}$ from the (potentially noisy) encoding $\mathbf{m}$ of the string returned from OCR of the scanned MRZ and the syndrome $\mathbf{s}$ decoded from the digital seal:

1. The syndrome $\mathbf{s}'$ of $\mathbf{m}'$ is computed.
2. From $\mathbf{v} = \mathbf{s} - \mathbf{s}'$, the error vector $\mathbf{e}$ is computed, corresponding to messages $\mathbf{y}$ having syndrome $\mathbf{v}$. The error vector of a message $\mathbf{y}$ is defined as the vector with minimum number of non-zero components (the error positions) so that $\mathbf{y} - \mathbf{e}$ is a codeword. Many Reed-Solomon decoders use a subroutine that computes the error vector from the syndrome.
3. The candidate value for $\mathbf{m}$ is computed as $\mathbf{m}' - \mathbf{e}$.

Provided that $\mathbf{m}$ and $\mathbf{m}'$ differ at most by $\lfloor (n-k)/2 \rfloor$ characters, the decoding returns the original MRZ content representation $\mathbf{m}$. The recovered vector is decoded, concatenated with the feature verification data and used as input for signature verification.

Any errors that have been corrected can be easily identified by the non-zero components of the error vector and should be displayed to a human operator for further inspection.

19.2.2 Facial Image Verification

The verification of a facial image must reliably detect replacement (substitutions) with an image of another person and should therefore also reliably detect replacement with a different image of the same person where the face might look different. Also, the process should be insensitive to wear-and-tear of the image due to normal use. Tamper detection of the face image is achieved by storing a limited number of essential features from the face image in the digital seal and later comparing them with the corresponding features extracted from the image on the document.

This approach raises the question of selecting relevant features from a facial image. For example, consider a digital representation of an image. The image can be transformed to the frequency domain by using a Fourier-transform resulting in the frequency components of the image. The frequency components with the highest magnitude represent important components for this particular image. Thus, the essential features of an image can be selected as the frequency components with the highest magnitude of the image. Comparing the most important frequency components of two images allows the verifier to decide if the frequency components (features) were generated from the same image.

19.2.3 Fingerprint Verification

Storing additional biometric information (e.g. fingerprint information) allows verifying the link between the document and the document holder. Since the data encoded in the digital seal is not encrypted, verification data in a digital seal for additional biometric feature should be stored as protected templates that do not reveal the biometric feature information. [FPTEMP] defines general requirements for methods to protect biometric information; a survey of biometric template protection techniques is given in [TEMPSURV].

An eligible method is the Fuzzy Commitment Scheme (FCS). The FCS can protect biometric information if this information is presented as an ordered set of features, called a feature vector. This approach presents the question of representing a fingerprint in the form of a feature vector. A possible approach for generating a feature vector for a fingerprint is to consider the fingerprint as a pattern of ridges. Several methods have

been published to represent patterns as vectors. For example, a feature vector can be constructed containing the direction of the ridges on the points of a grid around the core of a fingerprint. Different fingerprints have different ridge directions which makes it possible to distinguish fingers based on the generated feature vector. Such a feature vector can also be used as an input to the FCS, which takes care of protecting the contents of the feature vector and thus of the biometric information stored therein.

Annex A: Conversion of ECDSA Signature Formats (Informative)

The Conversion of ECDSA signature between ASN.1 and the raw format (r,s) is described in Annex B of [DOC 9303-13].

Annex B: C40-Encoding of Strings (Normative)

In order to save space in encoding alphanumeric characters and the filler symbol <, the encoding scheme C40 is used as defined in [DTMX]. In the following we define how these definitions are used in the current setting. The following two definitions apply to document features and their digital encoding:

1. Strings consist only of upper case letters, numbers, <SPACE>, and the symbol '<'. The latter is used as a filler symbol for the MRZ of travel documents. If '<' occurs in the string, all occurrences of '<' are replaced by <SPACE> before encoding. A string **MUST NOT** contain any other symbols.
2. Given a string of length L , the length (i.e. the number of bytes) of the corresponding digital encoding is the least even number that is larger or equal to L .

In the following calculations, we implicitly convert between a byte value and the corresponding unsigned integer equivalent. For example we define the value of a byte by a formula consisting of integer arithmetic on integer values.

Encoding

Encoding a string of characters into a sequence of bytes works as follows: First, the string is grouped into tuples of three characters, and each character is replaced with the corresponding C40 value according to Table 16, resulting in a triple $(U1, U2, U3)$. Then for each triple, the value

$$U = (1600 * U1) + (40 * U2) + U3 + 1$$

is computed. The result is in the range from 1 to 64000, giving an unsigned 16-bit integer value. This 16-bit value $I16$ is packed into two bytes by

$$\text{Byte } 1 = (I16) \text{ div } 256$$

$$\text{Byte } 2 = (I16) \text{ mod } 256$$

Here `div` denotes integer division (no remainder), and `mod` denotes the modulo operation. Note that these operations can be implemented by bit-shifting.

Decoding

The encoding can be easily inverted. Given a pair of bytes, let $(I1, I2)$ denote their unsigned integer values. The 16 bit value $I16$ is recalculated as

$$V16 = (I1 * 256) + I2$$

The triple $(U1, U2, U3)$ can be recomputed by

$$U1 = (V16 - 1) \text{ div } 1600$$

$$U2 = (V16 - (U1 * 1600) - 1) \text{ div } 40$$

$$U3 = V16 - (U1 * 1600) - (U2 * 40) - 1$$

Here again, `div` denotes integer division. Characters can be decoded from the triple $(U1, U2, U3)$ by simply looking up the corresponding values in Table 16.

Padding

The above definition is only well defined if the length of the string to be encoded is a multiple of three. Akin to the padding-definitions given in [DTMX], the following padding rules apply:

1. If two C40 values (meaning two characters) remain at the end of a string, these two C40 values are completed into a triple with the C40 value 0 (Shift 1). The triple is encoded as defined above.

2. If one C40 value (meaning one character) remains, then the first byte has the value 254dec (0xfe). The second byte is the value of the ASCII encoding scheme of DataMatrix of the character corresponding to the C40 value. Note that the ASCII encoding scheme in DataMatrix for an ASCII character in the range 0-127 is the ASCII character plus 1.

Example 1

Suppose the string “XK<CD” is to be encoded. By definition, all occurrences of '<' are replaced by <SPACE> before encoding. The resulting string is thus “XK CD”, i.e. “XK<SPACE>CD” (one space inserted). The C40 encoding/decoding of the string “XK<SPACE>CD” is depicted in Table 14.

Example 2

Suppose the “XKCD” is to be encoded. The string solely consists of uppercase letters. Its C40 encoding/decoding is depicted in Table 15.

Table 14: Encoding/Decoding example for the string “XK<SPACE>CD”.

Operation	Result			
original string	“XK<SPACE>CD”			
grouping into triples	(X, K, <SPACE>)	(C, D,)		
replacing with C40 values and padding	(37, 24, 3)	(16, 17, padding)		
calculating the 16 bit integer value	60164	26281		
	Byte 1 (div)	Byte 2 (mod)	Byte 1 (div)	Byte 2 (mod)
resulting byte sequence (decimal)	235	4	102	169
resulting byte sequence (hex)	0xeb	0x04	0x66	0xa9

Table 15: Encoding/Decoding example for the string “XKCD”.

Operation	Result			
original string	“XKCD”			
grouping into triples	(X, K, C)	(D, ,)		
replacing with C40 values and padding	(37, 24, 16)	(unlatch C40 and encode in ASCII)		
calculating the 16-bit integer value	60177			
	Byte 1 (div)	Byte 2 (mod)	Byte 1	Byte 2
resulting byte sequence (decimal)	235	17	254	69
resulting byte sequence (hex)	0xeb	0x11	0xfe	0x45

Table 16: C40 Encoding chart and correspondence to ASCII.

C40 Value	Character	ASCII Value	C40 Value	Character	ASCII Value
0	Shift 1	n/a	20	G	71

C40 Value	Character	ASCII Value	C40 Value	Character	ASCII Value
1	Shift 2	n/a	21	H	72
2	Shift 3	n/a	22	I	73
3	<SPACE>	32	23	J	74
4	0	48	24	K	75
5	1	49	25	L	76
6	2	50	26	M	77
7	3	51	27	N	78
8	4	52	28	O	79
9	5	53	29	P	80
10	6	54	30	Q	81
11	7	55	31	R	82
12	8	56	32	S	83
13	9	57	33	T	84
14	A	65	34	U	85
15	B	66	35	V	86
16	C	67	36	W	87
17	D	68	37	X	88
18	E	69	38	Y	89
19	F	70	39	Z	90

Annex C: Worked Example Arrival Attestation Document (Informative)

The following example gives an overview on how data are encoded into a digital seal according to the Arrival Attestation Document profile of Chapter 6. Figure 6 shows the front side of the Arrival Attestation Document with a DataMatrix code containing MRZ, AZR and a 64-byte elliptic curve signature, which is signed by the AAD-Signer.



Figure 6: Front side of the Arrival Attestation Document encoded in a DataMatrix bar code containing the MRZ, the AZR number and the elliptic curve signature

First, the whole input stream is given. This input stream consists of a header, a message and a signature zone and is now described in detail. The header starts with the magic constant 0xdc followed by the content version (0x02), the issuing country (D<<: Germany), the signer and certificate reference, where DE is the signer's country, TS identifies the signer and the last 5 characters (00027) are a reference of the certificate of the signer. The document is issued on the 1st January 2020 (0x0f7134) and the signature is created on 13th January 2020 (0x1145f4). The header ends with the Document Type Category (0x02) in the second line of the INPUT BYTESTREAM, identifying the document the seal is printed on.

The message zone contains the MRZ and AZR and begins with the tag 0x02 which refers to the MRZ (Machine-Readable-Zone) and its tag-length (0x30). After that, the content of the MRZ follows and consists of (see [MRTD6]):

1. Document Code and Issuing country (MED)
2. Surname (Mustermann) and first name (Erik) of document holder
3. Document number (M0000000)
4. Nationality (ALB) of the document holder
5. Date of birth of the document holder (YYMMDD: 030821)
6. Sex of document holder (M)

7. Date of expiry (160412).

After that, the AZR number is encoded. The tag is first encoded as 0x03 (AZR) and the tag-length is 0x08 which describes the length of the following information (160113000085). At the end, the signature is created (over header and message zone) and contains a marker 0xff (beginning of signature), the length (0x40) and the signature itself, (starts with 0x4a and ends with 0xe7).

INPUT BYTESTREAM:

```
-----
dc 02 6a bc 6d 32 c8 a5 19 fc 0f 71 34 11 45 f4
fd 02 02 30 a5 62 13 53 d9 a2 75 73 5b d4 13 4b
c5 49 13 3c 13 3c 13 3c 13 3c 13 3c a3 25 19 a5
19 a4 34 4a 5e 68 1a e7 20 4b 20 d5 32 cf 4b 7c
13 3c 13 3f 03 08 20 d5 20 10 19 a5 1a ea ff 40
4a 1f 21 8c a4 39 26 47 ec ff 6c 8a bf 9e 79 6a
78 ee be 0b 1a c8 cc 25 c4 ee 17 ee d9 61 d1 18
90 91 35 8d 7d 61 6f 1a 51 7a bc 74 7f 6c 44 90
ff 15 9d 4d cf 50 24 8b 00 b1 e3 2e 9e 78 05 e7
```

HEADER

```
-----
```

Magic Constant:

Hex: dc

Val: 220

Content Version:

Hex: 02

Val: 3 (according to [DOC 9303-13])

Issuing Country:

Hex: 6a bc

Val: D<<

Signer & Cert Ref:

Hex: 6d 32 c8 a5 19 fc

Val: DETS00027

Document Issue Date:

Hex: 0f 71 34

Val: 01/01/2020 (Jan 01st 2020)

Signature Creation Date:

Hex: 11 45 f4

Val: 01/13/2020 (Jan 13th 2020)

Feature Definition Reference:

Hex: fd

Val: 253

Annex D: Worked Example of a Social Insurance Card



*Figure 7: Digital Seal of a SIC containing:
social insurance number, first name, last
name, name at birth, issuing date and an
elliptic curve signature*

The following examples gives an overview how data are encoded in a digital seal according to the SIC profile of chapter 7. Figure 7 shows the digital seal of a SIC containing a social insurance card number, first- and last name and name of birth of the SIC owner, the issuing date of the SIC and an elliptic curve digital signature. The whole input stream (header, message zone and signature) encoded in the digital seal of the SIC is described in the following.

The header begins with a magic constant 0xdc (defines a bar code which has to be used in this specification), followed by the content version 0x02 (according to [DOC 9303-13]), the issuing country (D: Germany), the signer and certificate reference (DE identifies the signing country, TS defines the signer and the following 5 digit number (00027) is a reference to the specific certificate of the signer), the issuing date of the SIC (1 January 2020) and a 1-byte feature definition reference, which defines the data stored in the message zone. The 18-byte header ends with the Document Type Category (0x04) identifying the document the seal is printed on.

The encoding of the message zone starts with the tag 0x01 referring to the 8-byte (0x08) social insurance card number (65170839J003). The tag (0x02) describes the 11-byte length surname (Perschweiß), tag 0x03 defines the 5-byte length first name (Oscar) and tag 0x04 defines the name of birth (Jâcobénidictorius). After that, the signature is generated (over header and message zone) containing the tag 0xff (beginning of the signature) with a length (0x40) of 64 bytes and the signature itself. The described encoding of the raw input stream is shown below.

Raw Input ByteStream:

```
dc 02 6a bc 6d 32 c8 a5 19 fc 0f 71 34 11 6d 04
fc 04 01 08 3f ee 45 6d 2d e0 19 a8 02 0b 50 65
72 73 63 68 77 65 69 c3 9f 03 05 4f 73 63 61 72
04 13 4a c3 a2 63 6f 62 c3 a9 6e 69 64 69 63 74
75 72 69 75 73 ff 40 58 2e d0 5c b0 50 49 b4 fe
49 74 4d 5e 70 a1 98 2f 8b 7a fb 1e 90 25 9c e4
7b a9 90 7e fc f5 94 83 f0 a0 b4 21 ec 76 0d 6f
d2 28 4b ba c3 7e 13 6f 1e 3b e6 e1 4a 6f 6b c0
12 11 0c ff 91 9d 0f
```

HEADER

Magic Constant:

Hex: dc

Val: 220

Content Version:

Hex: 02

Val: 3 (indicating version 3 according to [DOC 9303-13])

Issuing Country:

Hex: 6a bc

Val: D<<

Signer & Cert Ref:

Hex: 6d 32 c8 a5 19 fc

Val: DETS00027

Document Issue Date:

Hex: 0f 71 34

Val: 01/01/2020 (Jan 01st 2020)

Signature Creation Date:

Hex: 11 6d 04

Val: 01/14/2020 (Jan 14th 2020)

Feature Definition Reference:

Hex: fc

Val: 252

Document Type Category:

Hex: 04

Val: 4

MESSAGE ZONE

Tag: 01 (Social Insurance Number)

Length: 08

Raw: 3f ee 45 6d 2d e0 19 a8

Val: 65170839J003

Tag: 02 (Surname)

Length: 11

Raw: 50 65 72 73 63 68 77 65 69 c3 9f

Val: Perschweiß

Tag: 03 (First name)

Length: 05

Raw: 4f 73 63 61 72

Val: Oscar

Tag: 04 (Name at Birth)

Length: 19

Raw: 4a c3 a2 63 6f 62 c3 a9 6e 69 64 69 63 74 75 72 69 75 73

Val: Jâcobénidicturius

SIGNATURE

Tag: 0xff

Signature Length: 64

Raw:

58 2e d0 5c b0 50 49 b4 fe 49 74 4d 5e 70 a1 98

2f 8b 7a fb 1e 90 25 9c e4 7b a9 90 7e fc f5 94

83 f0 a0 b4 21 ec 76 0d 6f d2 28 4b ba c3 7e 13

6f 1e 3b e6 e1 4a 6f 6b c0 12 11 0c ff 91 9d 0f

r:

58 2e d0 5c b0 50 49 b4 fe 49 74 4d 5e 70 a1 98

2f 8b 7a fb 1e 90 25 9c e4 7b a9 90 7e fc f5 94

s:

83 f0 a0 b4 21 ec 76 0d 6f d2 28 4b ba c3 7e 13

6f 1e 3b e6 e1 4a 6f 6b c0 12 11 0c ff 91 9d 0f

Annex E: Worked Example of a Digital Seal for Residence Permit Documents



Figure 8: Digital seal of a Residence Permit Document containing the MRZ and the passport number with an ECDSA brainpool 256 signature

The following worked examples shows how the data are encoded into a digital seal for the Residence Permit Document profile, described in section 8.4. The digital seal shown in Figure 8 is composed of a header, message zone and a signature. The header starts with the magic constant (0xDC) and is followed by the version (0x03), indicating version 4 according to [DOC 9303-13], the issuing country (D), the signer and certificate reference (DETS0227), where DE is signer's country and TS identifies the signer, 02 indicates the length of the following character (27), the issuing date (01.01.2020), the signature creation date, the Document Feature Definition Reference (0xfb), indicating the stored data in the bar code and the Document Type Category (0x06) identifying the document the seal is printed on.

After that, the message data are encoded. The message zone contains the MRZ and passport number and begins with the tag 0x02, which refers to the MRZ (Machine-Readable-Zone) and its tag-length (0x30). After that, the data of MRZ follows:

1. Document Code and Issuing Country (ATD)
2. Surname (Residorce) and firstname (Roland)
3. Document number (652584509)
4. Nationality (USA)
5. Date of birth (03 August 1970)
6. Sex (M)
7. Date of expiry (01 January 2022)

Next, the passport number is encoded. It starts with tag (0x03), followed by the length (0x06) and the encoded number (UFO001979). At the end, the signature is created (over header and message zone) and contains a marker 0xff (beginning of signature), the length (0x40) and the signature itself, (starts with 0x15 and ends with 0x27). In the following, the overall binary raw data encoded in the seal are shown.

Raw data encoded in the bar code:

```
dc 03 6a bc 6d 32 c8 a7 27 39 0f 71 34 11 45 f4
fb 06 02 30 5c 6a 13 58 75 97 6e c0 66 d4 17 b5
9e 8c 6a bc 13 3c 13 3c 13 3c 13 3c 3f ef 3a 29
38 ee 43 f1 59 3d 1a e5 2d bb 26 75 1f e6 4b 7c
```

13 3c 13 6b 03 06 d7 95 19 a6 53 06 ff 40 15 8c
f5 b0 bb 7f bd a0 99 e8 e7 c9 4a be fc 96 77 b2
e7 1e 98 85 f5 cd 90 57 a8 bf cf 93 0a 60 46 bc
f8 97 65 0a f8 f9 b0 34 3c 03 4d 2f 64 ce 08 1f
95 fb f3 d0 12 92 d1 f8 16 5e ce 27 d3 27

HEADER

Magic Constant:

Hex: dc

Val: 220

Content Version:

Hex: 03

Val: 4 (version 4 according to [DOC 9303-13])

Issuing Country:

Hex: 6a bc

Val: D<<

Signer & Cert Ref:

Hex: 6d 32 c8 a7 27 39

Val: DETS0227

Document Issue Date:

Hex: 0f 71 34

Val: 01/01/2020 (Jan 01st 2020)

Signature Creation Date:

Hex: 11 45 f4

Val: 01/13/2020 (Jan 13th 2020)

Feature Definition Reference:

Hex: fb

Val: 251

Document Type Category:

Hex: 06

Val: 6

MESSAGE ZONE

Tag:0x02(Machine Readable Zone)

Tag Length: 48 bytes

Hex: 5c ba 13 58 75 97 6e c0 66 d4 17 b5 9e 8c 6a bc
13 3c 13 3c 13 3c 13 3c 3f ef 3a 29 38 ee 43 f1

59 3d 1a e5 2d bb 26 75 1f e6 4b 7c 13 3c 13 6b

Val: ATD<<RESIDORCE<<ROLAND<<<<<<<<<<<
6525845096USA7008038M2201018<<<<<<06

Tag: 0x03 (passport number)

Tag Length: 6 bytes

Hex: d7 95 19 a6 53 06

Val: UF0001979

SIGNATURE

.....

Tag: 0xff (Signature marker)

Length: 64 bytes

RAW:

```
4a 8c e6 d8 ed ae bc 03 bc d2 62 c2 7c 7f d1 53
53 f6 85 71 17 00 d1 52 d6 59 82 57 18 5b d0 58
6a a3 24 a6 b7 78 90 e8 2f 14 54 95 e5 3f 64 f5
7b 6a 13 28 0f 0e cb 58 e6 4e bc 4f 37 b4 df c3
```

ri

```
4a 8c e6 d8 ed ae bc 03 bc d2 62 c2 7c 7f d1 53
53 f6 85 71 17 00 d1 52 d6 59 82 57 18 5b d0 58
```

S:

```
6a a3 24 a6 b7 78 90 e8 2f 14 54 95 e5 3f 64 f5
7b 6a 13 28 0f 0e cb 58 e6 4e bc 4f 37 b4 df c3
```

Annex F: Worked Example for a Digital Seal for VISA Stickers

The following worked example demonstrates how data are encoded into the digital seal for VISA stickers as described in section 5.



Figure 9: Digital Seal for VISA sticker containing MRZ, Duration of Stay and passport number signed with brainpoolP224r1 curve

The digital seal as shown in Figure 9 is composed of a header, the message zone and the signature. The header according to [DOC 9303-13] starts with the magic constant (0xDC), followed by the version (0x03), i.e. version 4 according to [DOC 9303-13], the issuing country (D<<), the signer and certificate reference (DETS0232), where DE is the signer's country and TS identifies the signer, 02 indicates the length of the following character (32), the issuing date (01.01.2020), the signature creation date, the Document Feature Definition Reference (0x5D), indicating the stored data in the bar code and the Document Type Category (0x01), identifying the document the seal is printed on.

The message zone consists of the MRZ of type B, the duration of stay and the number of the passport the VISA is attached to. The MRZ of type B is identified by tag (0x02) and has a length of 44 bytes (0x2C). The MRZ data is:

1. Document Code and Issuing Country (VCD)
2. Surname (DENT) and first name (ARTHUR PHILIP)
3. Document number (1234567XY)
4. Nationality (GBR)
5. Date of birth (11 March 1952)
6. Sex (M)
7. Date of expiry (25 May 2020)

After that, the encoding of the duration of stay follows. It is identified by tag number (0x04) with a length of 3 bytes (0x03) for the following data (0xA00000). The last data of the payload encode the passport number identified by tag number (0x05) following by a length of (0x06) bytes and the passport number itself (0x33be1fed20c6=47110815P).

The generated signature starts with tag (0xff) followed by a length of (0x38) bytes and the signature itself.

Raw data encoded in the seal (see Figure 9):

```
dc 03 6a bc 6d 32 c8 a7 2c b1 0f 71 34 b7 98 15
5d 01 02 2c dd 52 13 4a 74 da 13 47 c6 fe d9 5c
```

b8 9f 9f ce 13 3c 13 3c 13 3c 13 3c 20 38 33 73
4a af 47 f0 c3 2f 1a 1e 20 eb 26 25 39 3a fe 31
04 03 a0 00 00 05 06 33 be 1f ed 20 c6 ff 38 d5
57 32 47 60 8c 9b 64 ca e0 22 c1 c5 24 96 60 e5
1a e3 16 d6 00 f9 74 aa c2 12 9c 3f 09 3c 35 95
53 14 02 63 04 70 5c d2 4c 1b 81 00 71 1b d4 41
1f 06 8e 8d 93 e8 b7

HEADER

Magic Constant:

Hex: dc

Val: 220

Content Version:

Hex: 03

Val: 4 (version 4 according to [DOC 9303-13])

Issuing Country:

Hex: 6a bc

Val: D<<

Signer & Cert Ref:

Hex: 6d 32 c8 a7 2c b1

Val: DETS0232

Document Issue Date:

Hex: 0f 71 34

Val: 01/01/2020 (Jan 01st 2020)

Signature Creation Date:

Hex: b7 98 15

Val: 12/03/2021 (Dec 03rd 2021)

Feature Definition Reference:

Hex: 5d

Val: 93

Document Type Category:

Hex: 01

Val: 1

MESSAGE ZONE

Tag:0x02(Machine Readable Zone)

Tag Length: 44 bytes

Annex G: Worked Example for the Address Sticker for the German ID-Document

The following worked example demonstrates how the data for the address sticker for German ID-Documents are encoded in a digital seal as described in section 9.



Figure 10: Digital Seal for an address sticker containing the passport number, official municipality code number and residential address signed with brainpoolP224r1 curve

The digital seal for the address sticker for the German ID document is composed of a header, the payload and the signature securing the data stored in a seal against manipulations. The header according to [DOC 9303-13] starts with the magic constant (0xDC), followed by the version number (0x03), i.e. version 4 according to [DOC 9303-13], the issuing country (D<<), the signer and certificate reference (DETS0232), where DE is signer's country and TS identifies the signer, 02 indicates the length of the following character (32), the issuing date (01.01.2021), the signature creation date (03.12.2021), the Document Feature Definition Reference (0xF9) and the Document Type Category (0x08) identifying the document the digital seal is printed on.

The message zone consists of the document number the sticker is attached to, the official municipality code number and the holder's residential address, composed of the concatenation of the postal code, the name of the street and the house number. The document number starts with the tag (0x01) and has a length of 6 bytes (0x06). The document number is T2000AK47 and is C40-encoded as 0xcf3519af974c. After the document number, the official municipality code number follows with tag 0x02 and is encoded in a length of 6 bytes (0x06). The official municipality code number in the worked example is (05314000), which stands for the city "Bonn" and is encoded as 0x1a70208519a1. The residential address has the tag 0x03, a length of 16 bytes (0x10) and contains the value 53123MUSTERMANNSTRASSE21, encoded as 0x395e26b3d9a275735bd4cd485ca17176. The generated signature starts with tag (0xff), followed by a length of (0x38) bytes and the signature itself.

The raw data encoded in the digital seal according to Figure 10 is shown below:

```
dc 03 6a bc 6d 32 c8 a7 2c b1 0f 71 35 b7 98 15
f9 08 01 06 cf 35 19 af 97 4c 02 06 1a 70 20 85
19 a1 03 10 39 5e 26 b3 d9 a2 75 73 5b d4 cd 48
5c a1 71 76 ff 38 91 01 f3 05 c9 aa 04 d2 ab 00
21 2c 7c f6 5f ef ca af 56 5a 54 53 8b 24 98 d6
81 a1 88 39 c9 26 56 52 6c 40 59 4b 0b 1d a1 2a
```

27 f6 87 a3 50 12 78 dc 41 45 b6 e6 4a fc

HEADER

Magic Constant:

Hex: dc

Val: 220

Content Version:

Hex: 03

Val: 4 (version 4 according to ICAO)

Issuing Country:

Hex: 6a bc

Val: D<<

Signer & Cert Ref:

Hex: 6d 32 c8 a7 2c b1

Val: DETS0232

Document Issue Date:

Hex: 0f 71 35

Val: 01/01/2021 (Jan 01st 2021)

Signature Creation Date:

Hex: b7 98 15

Val: 12/03/2021 (Dec 03rd 2021)

Feature Definition Reference:

Hex: f9

Val: 249

Document Type Category:

Hex: 08

Val: 8

MESSAGE ZONE

Tag:0x01(Document Number)

Tag length: 0x06

Hex: cf 35 19 af 97 4c

Val: T2000AK47

Tag: 0x02 (Official Municipality Code Number)

Tag length: 0x06

Hex: 1a 70 20 85 19 a1

Val: 05314000

Tag: 0x03 (Residential Address)

Tag length: 0x10

Hex: 39 5e 26 b3 d9 a2 75 73 5b d4 cd 48 5c a1 71 76

Val: 53123MUSTERMANNSTRASSE21

SIGNATURE ZONE

Tag: 0xFF (Signature)

Tag length: 0x38

RAW:

91 01 f3 05 c9 aa 04 d2 ab 00 21 2c 7c f6 5f ef ca af 56 5a 54 53 8b
24 98 d6 81 a1 88 39 c9 26 56 52 6c 40 59 4b 0b 1d a1 2a 27 f6 87 a3
50 12 78 dc 41 45 b6 e6 4a fc

r: 91 01 f3 05 c9 aa 04 d2 ab 00 21 2c 7c f6 5f ef ca af 56 5a 54 53 8b
24 98 d6 81 a1

s: 88 39 c9 26 56 52 6c 40 59 4b 0b 1d a1 2a 27 f6 87 a3 50 12 78 dc 41
45 b6 e6 4a fc

Annex H: Worked Example for the Residence Sticker for German Passports

The following worked example shows how the data for the residence sticker for German passports are encoded in the digital seal according to the profile described in section 10.



Figure 11: Digital seal for the residence sticker for German passports containing the passport number, municipality code number and the postal code signed by a brainpoolP224r1 curve

The digital seal for the residence sticker of the German passport is composed of a header, payload and the signature, securing the seal against manipulation. The header as described in [DOC 9303-13] starts with the magic constant (0xDC), followed by the version number (0x03), i.e. version 4 according to [DOC 9303-13], the issuing country (D<<), the signer and certificate reference (DETS0232), where DE is the signer's country and TS identifies the signer, 02 indicates the length of the following character (32), the issuing date (01.01.2021), the signature creation date (03.12.2021), the Document Feature Definition Reference (0xF8) and the Document Type Category (0x0A) identifying the document the digital seal is printed on.

After the header, the message zone follows and is made up of the number of the passport the sticker is attached to, the municipality code number of the issuing authority and the postal code of the domicile of the passport holder. The document number starts with the tag (0x01) with a length of 6 bytes (0x06) and its number PA5500K11 C40-encoded as (0xb77a38e596ce). After the document number, the municipality code number is followed by tag 0x02 with a length of 6 bytes (0x06) and the value (03359010) of the issuing authority, which stands for Buxtehude and is C40-encoded as (0x1a203a4d1fe1). The postal number starts with tag 0x03 and has a length of 4 bytes. The postal code of 21614 is C40-encoded as 0x26532081. The signature zone starts with the tag 0xff, followed by a length of 56 bytes (0x38) and the signature itself.

The complete raw data of the residence sticker depicted in Figure 11 is shown below:

```
dc 03 6a bc 6d 32 c8 a7 2c b1 0f 71 35 b7 98 15
f8 0a 01 06 b7 7a 38 e5 96 ce 02 06 1a 20 3a 4d
1f e1 03 04 26 53 20 81 ff 38 14 62 bc fe f5 1a
d1 b8 d0 9d e3 28 b1 5c 54 35 e6 48 29 ce 54 86
47 dd 57 97 87 16 96 24 a0 46 73 8d c5 02 37 08
31 9a 42 94 81 11 31 ea ac 58 d5 18 cf e9 8b 43
14 37
```

HEADER

Magic Constant:

Hex: dc

Val: 220

Content Version:

Hex: 03

Val: 4 (version 4 according to ICAO)

Issuing Country:

Hex: 6a bc

Val: D<<

Signer & Cert Ref:

Hex: 6d 32 c8 a7 2c b1

Val: DETS0232

Document Issue Date:

Hex: 0f 71 35

Val: 01/01/2021 (Jan 01st 2021)

Signature Creation Date:

Hex: b7 98 15

Val: 12/03/2021 (Dec 03rd 2021)

Feature Definition Reference:

Hex: f8

Val: 248

Document Type Category:

Hex: 0a

Val: 10

MESSAGE ZONE

Tag:0x01(Document Number)

Tag length: 0x06

Hex: b7 7a 38 e5 96 ce

Val: PA5500K11

Tag: 0x02 (Official Municipality Code Number)

Tag length: 0x06

Hex:1a 20 3a 4d 1f e1

Val: 03359010

Tag: 0x03 (Postal Code)

Tag length: 0x04

Hex: 26 53 20 81

Val: 21614

Tag: 0xFF (Signature)

Tag length: 0x38

RAW: 14 62 bc fe f5 1a d1 b8 d0 9d e3 28 b1 5c 54 35 e6 48 29 ce 54 86 47
dd 57 97 87 16 96 24 a0 46 73 8d c5 02 37 08 31 9a 42 94 81 11 31 ea
ac 58 d5 18 cf e9 8b 43 14 37

Val: 14 62 bc fe f5 1a d1 b8 d0 9d e3 28 b1 5c 54 35 e6 48 29 ce 54 86
47 dd 57 97 87 16
96 24 a0 46 73 8d c5 02 37 08 31 9a 42 94 81 11 31 ea ac 58 d5 18
cf e9 8b 43 14 37

Annex I: Worked Example for Temporary or Substitutional Travel and ID Documents

The following worked example describes, how the data of the temporary passport are encoded in the digital seal for one of the profiles defined in of chapter 12. It has to be noted, that only one worked example for the defined profiles according to chapter 12 to chapter 17 is given, because they all follow the same encoding and only differ in the data of the message zone.

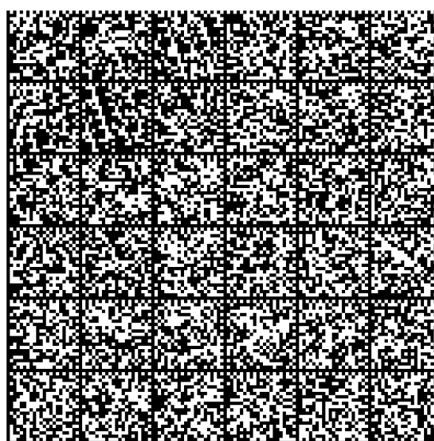


Figure 12: Worked Example for a Temporary Passport according to chapter 12

The digital seal shown in Figure 12 is composed of a header, a message (encoding the payload data) and signature zone according to [ICAO TR-IDB]. These data are then encoded in base 32 appended to the prefixes of Bar Code Identifier (“RDB1”) and the Bar Code Flag (“B”) and stored in the digital seal. The detailed encoding is described in the following:

The header as described in [ICAO TR-IDB] starts with the Country Identifier (“D<<”) encoded in two bytes using C40-encoding (0x6abc), followed by the signature algorithm “ECDSA with SHA-256” encoded in one byte as 0x01, the Certificate Reference identifying the signer certificate used for signature verification, encoded by the last 5 bytes of SHA-1 of the signer certificate (0xa57a790577) and the Signature Creation Date “June 10th 2026” expressed in 4 bytes (0x005D1C0A). Thus the header has a fixed length of 12 bytes.

After the header, the message zone follows and start with tag (0x61) identifying the Bar Code Information Group Template (see [ICAO TR-IDB]) with a length encoding of three bytes (0x820431), where the first byte of the length encoding (0x82) signifies that a 2-byte length (0x0431) are follows and thus defining the overall length of the encoded data in the Bar Code Information Group Template. The first message type encoded in the bar code starts with tag (0x08) identifying the MRZ-TD3 as defined in section 11.2.2, followed by the length encoding (0x3c). After the length encoding, the MRZ-TD3 data follow C40-encoded. After the MRZ, the face image is encoded and starts with tag (0x80) following by three byte length encoding (0x8203e6). Again, the first byte (0x82) of the length encoding signifies that a 2-byte length (0x03e6) follows, defining the number of bytes encoding the face image followed by the data bytes of the face image themselves. After the face image, the expiration date of the document is encoded by tag (0x84) of 4 bytes length. The expiration date has the value (0x006F929B), where the first byte is the date mask (see section 11.2.2) and the remaining three bytes (0x6F929B) are encode the expiration date (July 31st 2027). The last message type of the message zone is the national document type identifier encoded by tag (0x86) with one byte length (0x01) and the value (0x06), which is the identifier for the Temporary Passport as defined in Table 8.

The signature zone starts with tag (0x7f) with a length of 64 bytes (0x40) and the value contains the signature itself.

The complete byte stream for the temporary passport encoded in the digital seal according to Figure 12 is shown below:

```
6abc01a57a790577005D1C0A61820431083cb9da134cb301c87cd08aa1aa5acc133c133c133c
133c133c133c133c134020104ce833ee6abc52fd52123b3e3f2619d0133c133c133c133c133d
fe35808203e60000000c6a5020200d0a870a00000014667479706a7032200000000006a703220
000000476a7032680000000166968647200000008100000064000307070000000000f636f6c72
010000000000100000001a726573200000001272657363012c00fe012c00fe0404000000006a
703263ff4fff51002f0000000000640000000810000000000000000000064000000081000000
0000000000003070101070101070101ff5c00174267355ecd5ecd5ee432a932a933012a862a
862a3bfff52000c00000001010304040000ff64000e00014c545f4a50325f323236ff90000a00
000000030b0001ff93cfae9a27fe59049f42ca97bf8d628b9c3b5b945b9ff5ac07ad474890df
90e311a6bf98b745c1ba370e99271b5f18060642f4c41cd97d553fc2ec896689b51b45b79f44
9440e3f7f3ef9a307476106cdcc7ff15dcdd1542ebb9444158f2560742cec1fcf9d9dbbbbd031
c9f4bdc024a1098687e2872f1eb21095656e4221721cef9eb00552b7f5c0c504c3ee5b995757
c8d92aad7f2f31ecc96abfc003ee8080c3e5adc3e6b541f1d34080616250b1a2225ef5d1d4af
668417ef60b2f94c9cd7c8c1bcafe885ff0048155eec769683a02a3ebb6e7330f1ae5fa075f0
f93adfb504433c978f3cd3164aff5461a628540172aa19a081b12fc48092ae282be8da46cfa8
72c6296190dd036db1597dbbab513ba2c0dd91562fe07e8e804873e0f9e90a8063b25f00e885
9b95075302012de14fa5810041a8adf489bd7fd3ad1c43609fd2e67dbd299a024b6a1716e3a9
1178d2a5e4876dcb871826becd2f72d6264d7fd925747d7775f223597cb17b5365582a2c8c5d
8bbb0e6a171b4692bb0f5a89ae5f7bdf4e375f63f38c5e85d9188c46dc4c7907d22a0e5096d7
ea27cd6402dc6525f12d1900770461e1dbd986d7ae45dcce479b64e9ad9f7d03ab2d8080c7c7
a123e3bbclcd48c32140f1dfc72ec1dfd16bea407ce5829f6a02c4ba1bc9251deff7f691fed6
7c8618e2b714a57d4ab06002d2207e5ac7d6f39f761e6e519898f04e4353621f5bc91ef4778b
73c1870983d347ad2d037c877922d747448e334d6f94ea1696acae4c3d975a30e2fc5fef5640
7a4265272145c6fed923f6cb629fbbb4c546d83d583d098722c32cff25d58eac2d91fa3a655d
4f172df4898cc5ca150e82de9e46ba8be87b1f005b182cb7b8a1b9c95697d47a956daceb9c08
62e98bc62182e5df3387c4d2c559db5da0afa761495c9ef3b0d4eb4522a7205e1eb084b94fc5
238d01580296f6f799d94decc9050406898eeec95b6361fa309ef2eca3c2bd9b9255a593a0f0
3017951b7b26bfa52cdf7c3bd84e29386d6823c5f41d24d04d398d0ce5752b82398080a1c600
ea4f41509a32ce8685bf0a738080ffd98404006F929B8601067f40956f3405391530f49b2774
626a0e4a3021a3e3162bff321f5cc0e831b6d9c920721794c05b002332647443a5fe0f16b565
47cccd4871ea990b377543026a8cdb
```

HEADER

Country Identifier:

Hex: 0x6abc

Val: D<<

Signature Algorithm:

Hex: 0x01

Val: ECDSA-SHA-256

Certificate Reference:

Hex: 0xa57a790577

Val: last 5-bytes of SHA-1(Certificate: UTTS5B)

Tag:0x84(Expiration Date)

Length: 0x04

Hex: 0x006F929B

Val: 07/31/2027 (July 31st 2027)

Tag:0x86 (National Document Type Identifier)

Length:0x01

Hex: 0x06

Val: 06 (Temporary Passport)

SIGNATURE ZONE

Tag:0x7F (Signature tag)

Length: 0x40

Hex: 95 6f 34 05 39 15 30 f4 9b 27 74 62 6a 0e 4a 30 21 a3 e3 16 2b ff 32 1f
5c c0 e8 31 b6 d9 c9 20 72 17 94 c0 5b 00 23 32 64 74 43 a5 fe 0f 16 b5 65
47 cc cd 48 71 ea 99 0b 37 75 43 02 6a 8c db

Val: Signature

BASE32 (with RDB1 prefix), encoded in the bar code :

RDB1BNK6ADJL2PECX0AC5DQFGDAQEGEEDZ002CNGLGA0IPTIIVINKLLGBGPATHQJTIEZ4CM6BGPA
THQJUAIAQJTUDH3TKXRJP2UQSHM7D6JQZ2AJTYEZ4CM6BGPATHX7DLAECAPTAAAAABRVFAIBABUF
I0CQAAAABIZTUPFYGU4BSEAAAAAANJYDEIAAAAAE02TQGJUAAAAACZUWQZDSAAAAABAAAAAGIAA
DA4DQAAAAAAA6Y3PNRZACAAAAAAAEEAAAAABU4TF0MQAAAAACJZGK43DAEWAB7QBFQAP4BAEAAA
AAADK0AZGH72P75IQALYAAAAAADEAAAAABIAAAAAAAAAAAAAAAAAAMQAAAAEBAAAAAAAAAAAAAAA
DA4AQCBYBAEDQCAP7LQAB0QTHGVPM2XWNL3SDFKJSVEZQCKUGFKDCU077KIAAYAAAAAAQCAYEAQA
AB73EAAHAAAKMKRPUUUBSL4ZDENX7SAAUAAAAAAAGCYAAH7ZHT50TIT74WIET5BMVF57RVRIXHB
3LOKFHXH7VVD22R2ISDPZBYRU27ZRN2FYG5D0DUZE4NV6GAGAZBPJRA43F6VKP6C5SEWNCNVDNC
3PH2ESRA0H57T56NDA5DWCBNZR77CX0N2FKC504UIQKY6JLA0QW0YH6PTXN3XPIDDSPUXXACJII
JQ2D6FBZPD2ZBBFLFNZBCC4Q456PLABKSW724BRIEYPXFXGKXK7ENSKVNP4XTD3GJNK74AA70QCA
MHZNNYPTLKQPR2NAIAYLCKCY2EIS66XI5JL3GQQL66YFS7FGJZV6IYG6K72EF74AEQFK65R3JNA5
AFI7LW3TTGDY24X5A0XYP50W7WUCEGPEXR46NGFSK75KGDJRIKQAXFKQZUCA3CL6EQCJ4KBL5DN
ENT5I0LDCSYM3UBW3MKZPW52WUJ3ULAN3EKWF7QH5DUAJBZ6B6PJBKAGHMS7ADUJLG4VA5JQEAJ
N4FH2LAIAGUK35EJXV75HLI4INQJ7UXGPW6STGQCJNVB0FXDVEIXRUVF4SDW3S4HDATL5TJPOLL
CMTL73ESXI7LX0XZCWL4WF5VGZKYFIWIYXMLXMHGUFY3I2JLWD22RGXF6667JY3V6Y7TRRPILWI
YRRDNYTDZA7JCUDSQS3L6UJ6NMQBPNYJZF6EWR5AD5XARQ6DW6ZQ3L24R04ZZDZWZHVWPX2A5LFWA
IBR6HUER6H06BZVEMGIKA6HP40LWB37IWX2SAPTSYFH3KALCLUG6JEU06757WSH7NM7EGDDRL0FF
FPVFLAYAC2IQH4WWH23ZZ65Q6NZIZRGHQJZBVGQ7LPER55DXRNZ4DBYJQPJUPLJNAN6I06JC25D
UJDRTJVXZJ2QWS2WK4TB5S5NDBYX4L7XVMQD2IJS0IKFY37NSI7WZNRJ705UYVDNQPKYHUEY0IW
DFT7SLVM0VQWZD6R2MV0U6FZN6SEYZROKCUHIFXU6I25IX2D3D4AFWGBMW64KD00JK2L5I6UVNWW
OXHAIMLUYXRRBQLS56M4HYTJMKW03LWQK7J3BJF0J545Q2TVUKIVHEBPB5MEEXFH4KI4NAFMFFX
W66M5STPMZECQIBUJR3XMSW3DMH5DBHXS5SR4FPM3SJK2LE5A6AYBPFI3PMTL7JJM356DXWCOFE4
G22BDYX2B2JGQU4Y2DHF0UVYE0MAQCQ4MAHKJ5AVBGRSZ2DILPYK00AIB76ZQQCAA34STODACBT

7IBJZB0K6QZX7RUDKTIWXGPLQPDQKUGN6N6FWVK6VABMXUA6H6JPATAQVACQZTBBE3LSP3YC25N
ZYHFQ72YXIYX4HBXKQCGIYPTU5WwY

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